UNIVERSITÄT PASSAU

Software Analysis

Interprocedural and Heap Analysis

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Lehrstuhl für Software Engineering II

Outline

1. Limitations of Intraprocedural Analysis
2. Interprocedural Analysis
3. Dataflow Analysis and the Heap

Limitations of Intra-Procedural Analysis

```
int a = 7;
int d = f(a, 2);
int e = a + d;
```

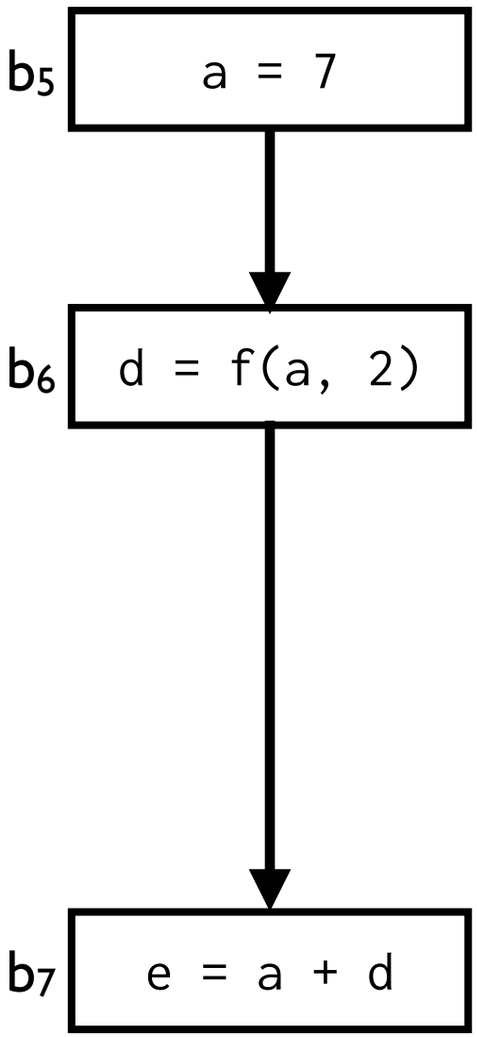
If you do not know anything about methods or functions the analysis must be conservative: it is $\top$ or $\bot$

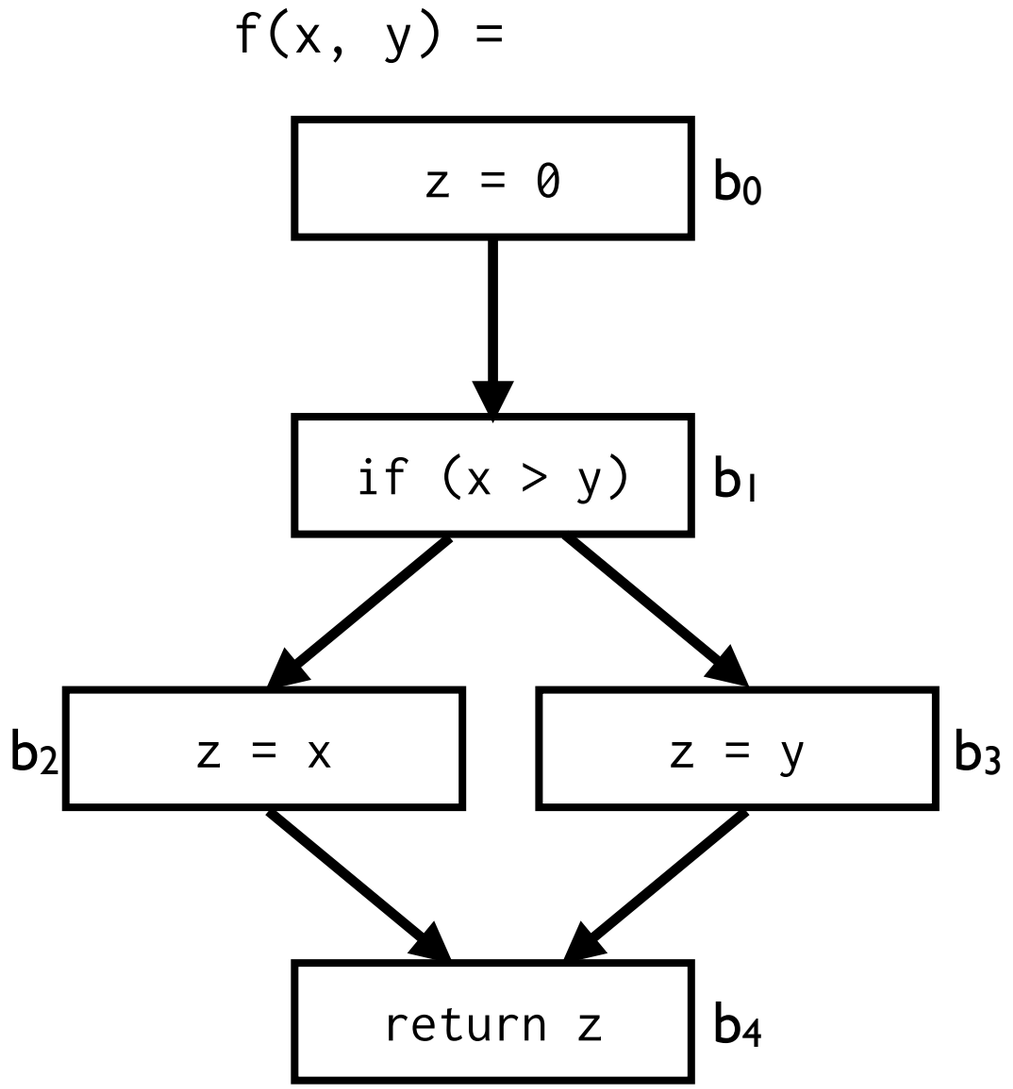
Limitations of Intra-Procedural Analysis

```
int a = 7;
int d = f(a, 2);
int e = a + d;
```

```
int f(int x, int y) {
  int z = 0;
  if (x > y) {
    z = x;
  } else {
    z = y;
  }
  return z;
}
```

Limitations of Intra-Procedural Analysis





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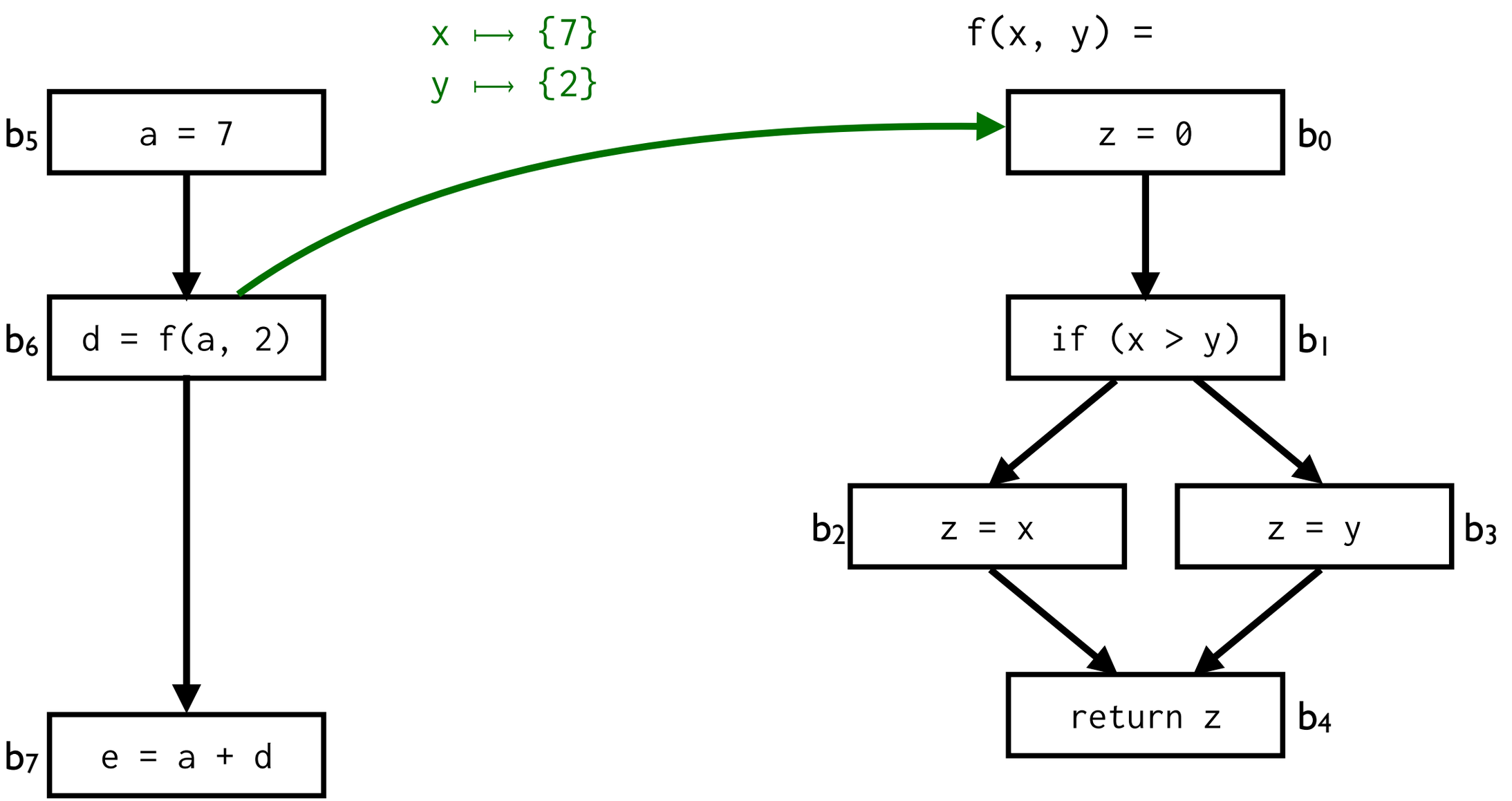

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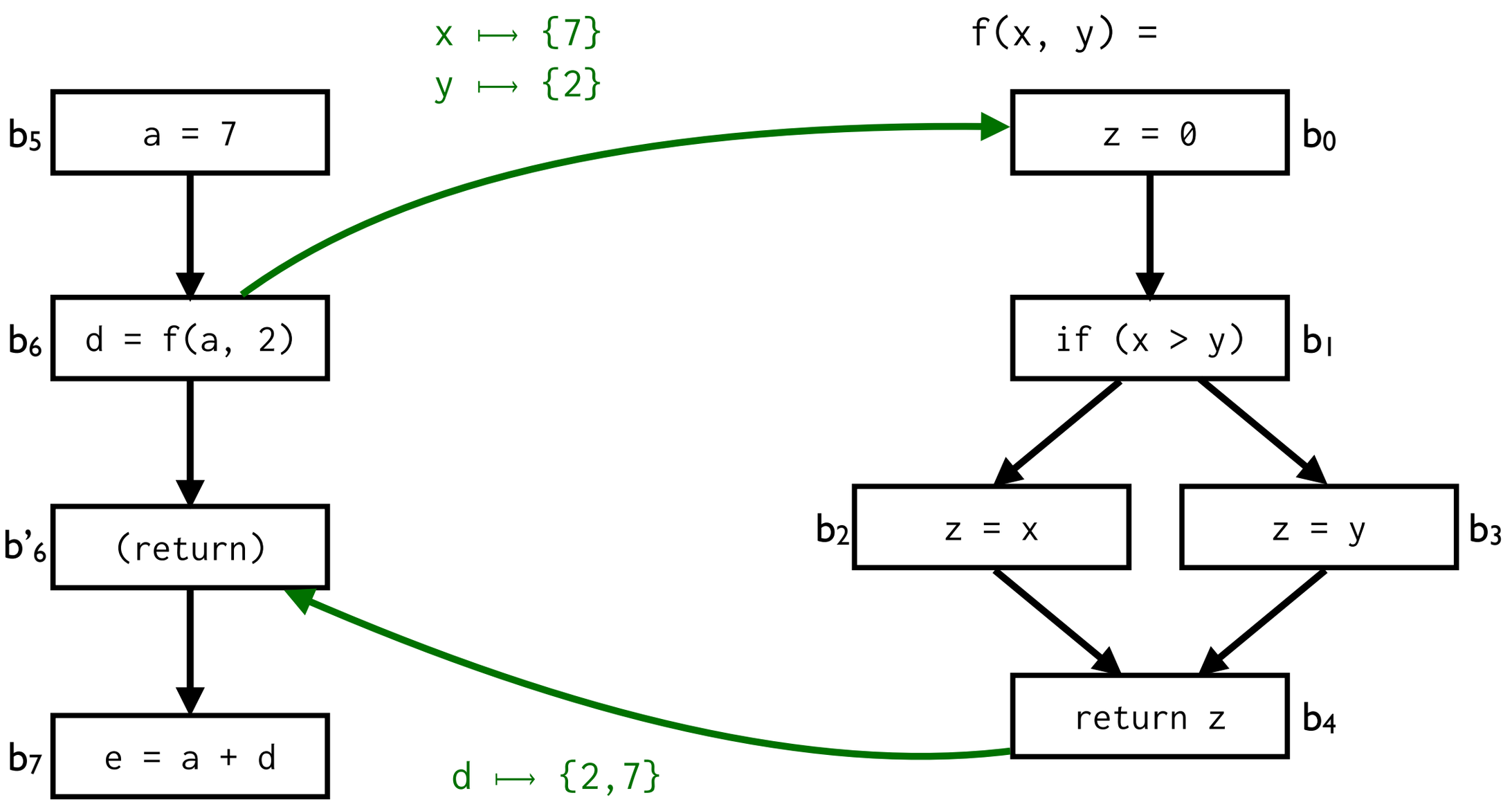

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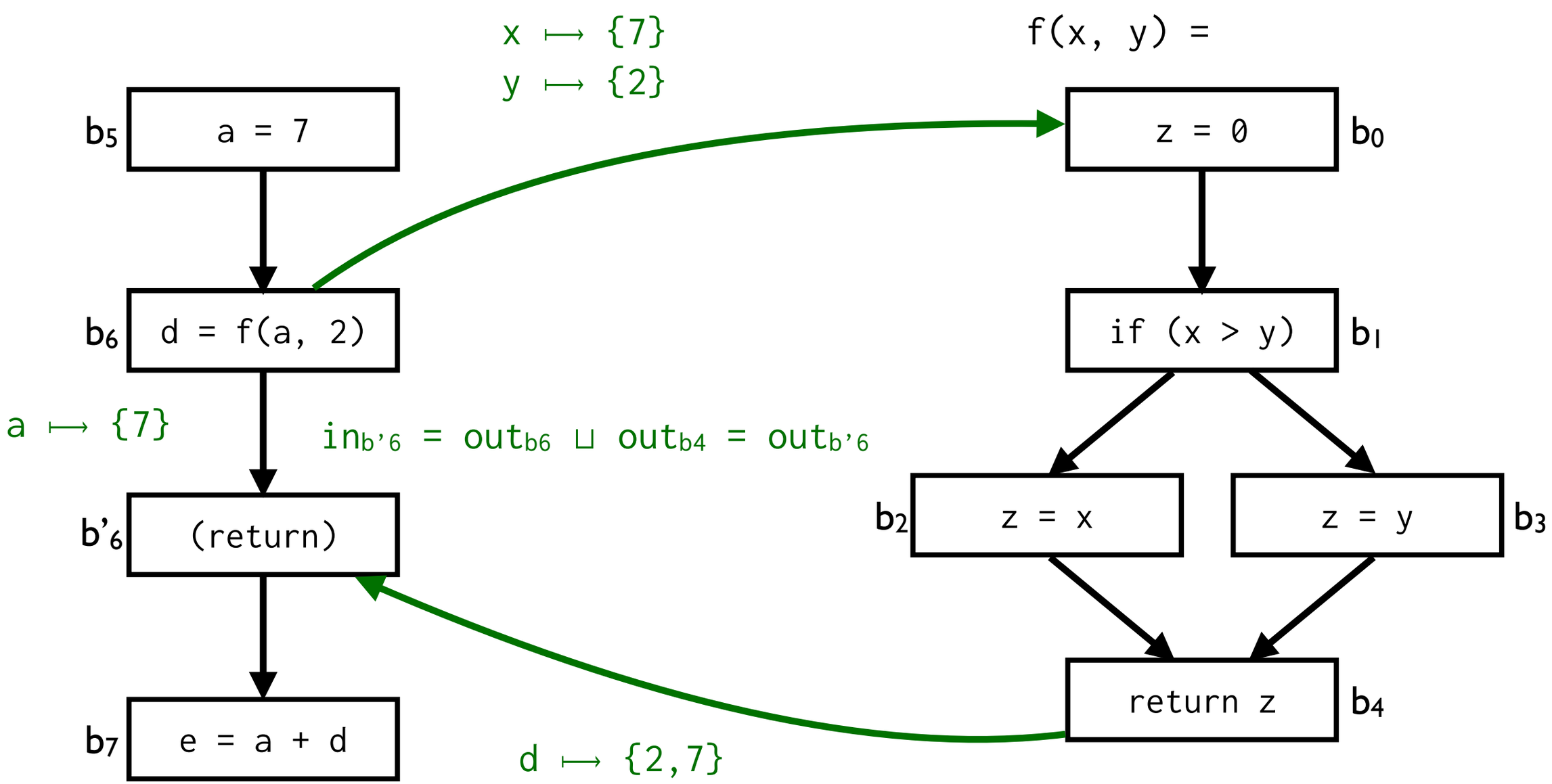

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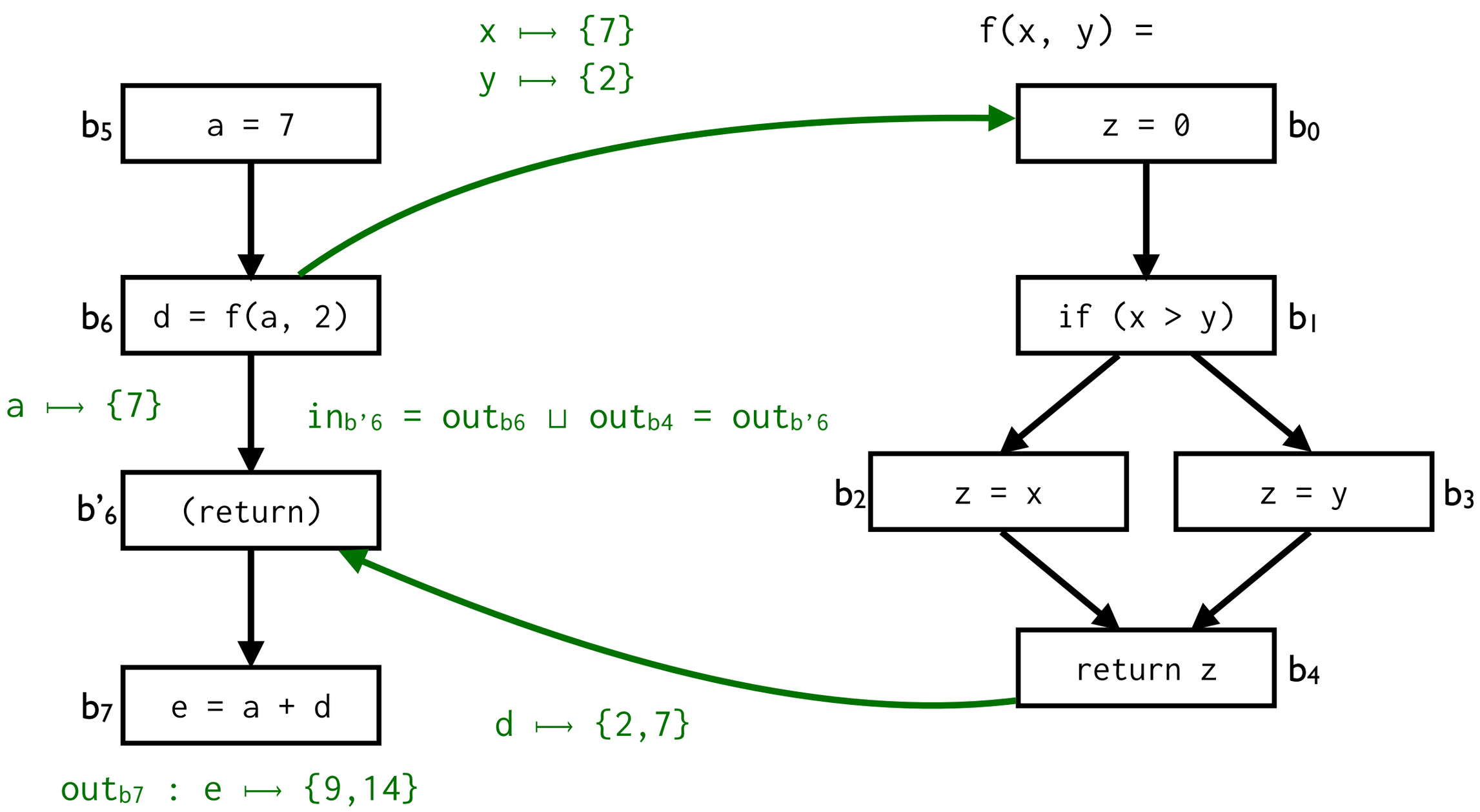

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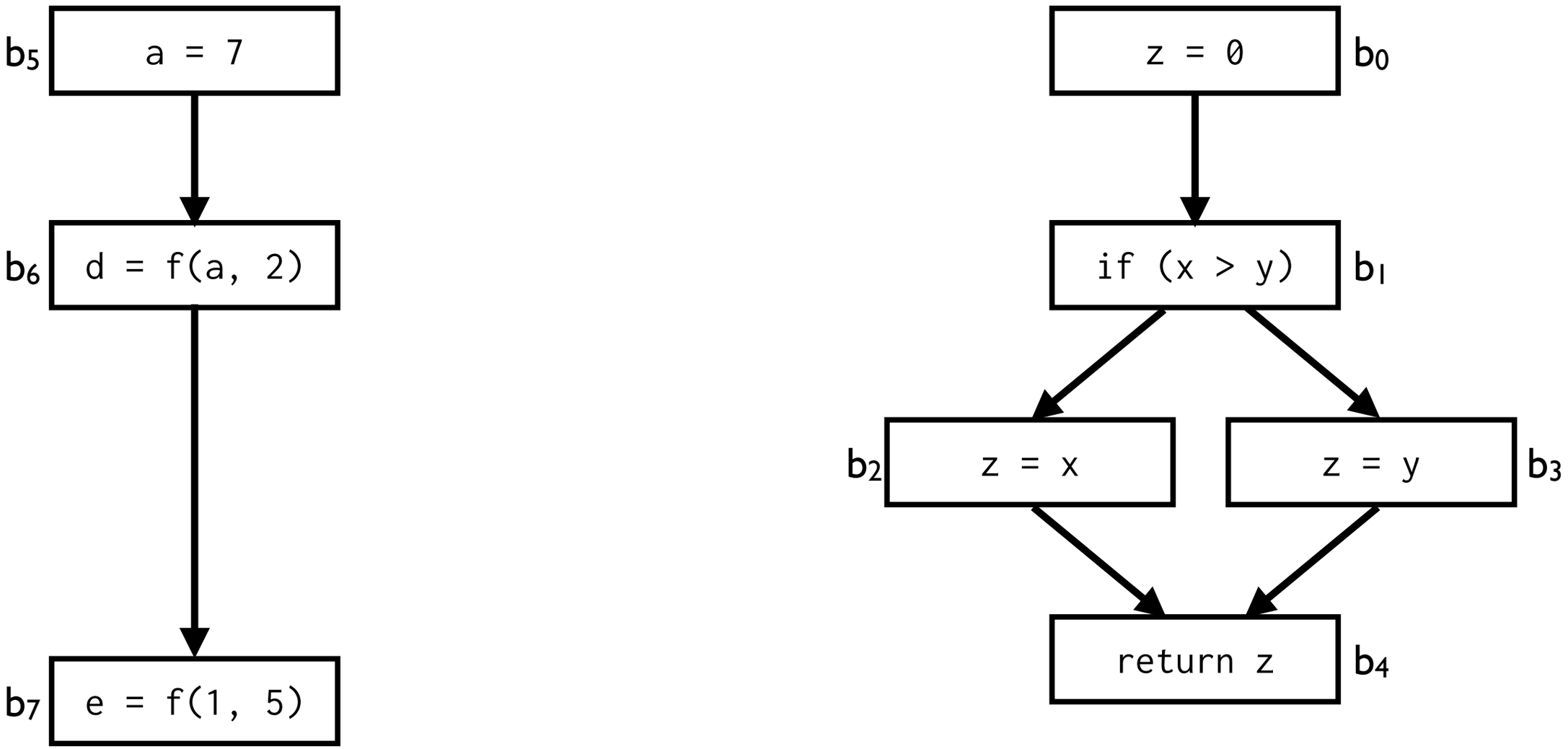

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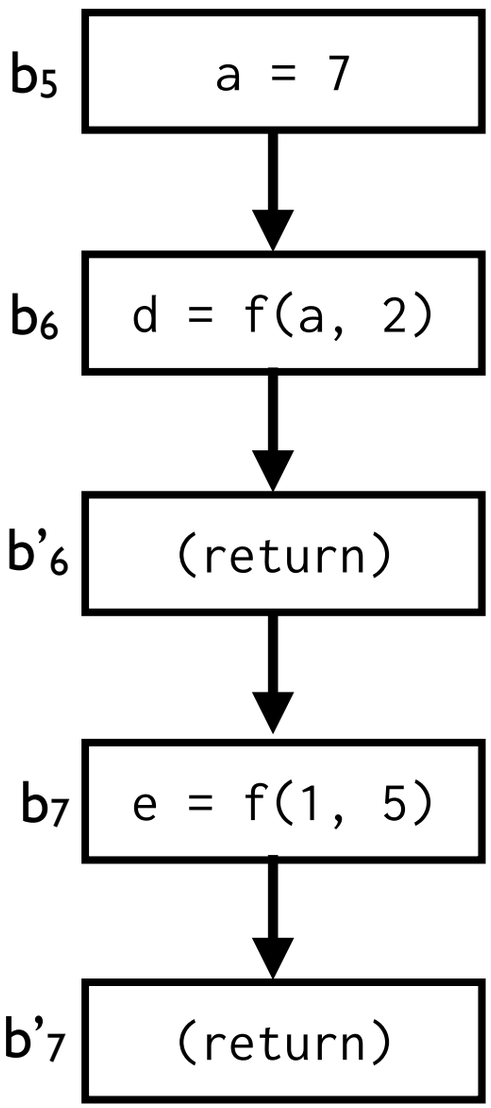

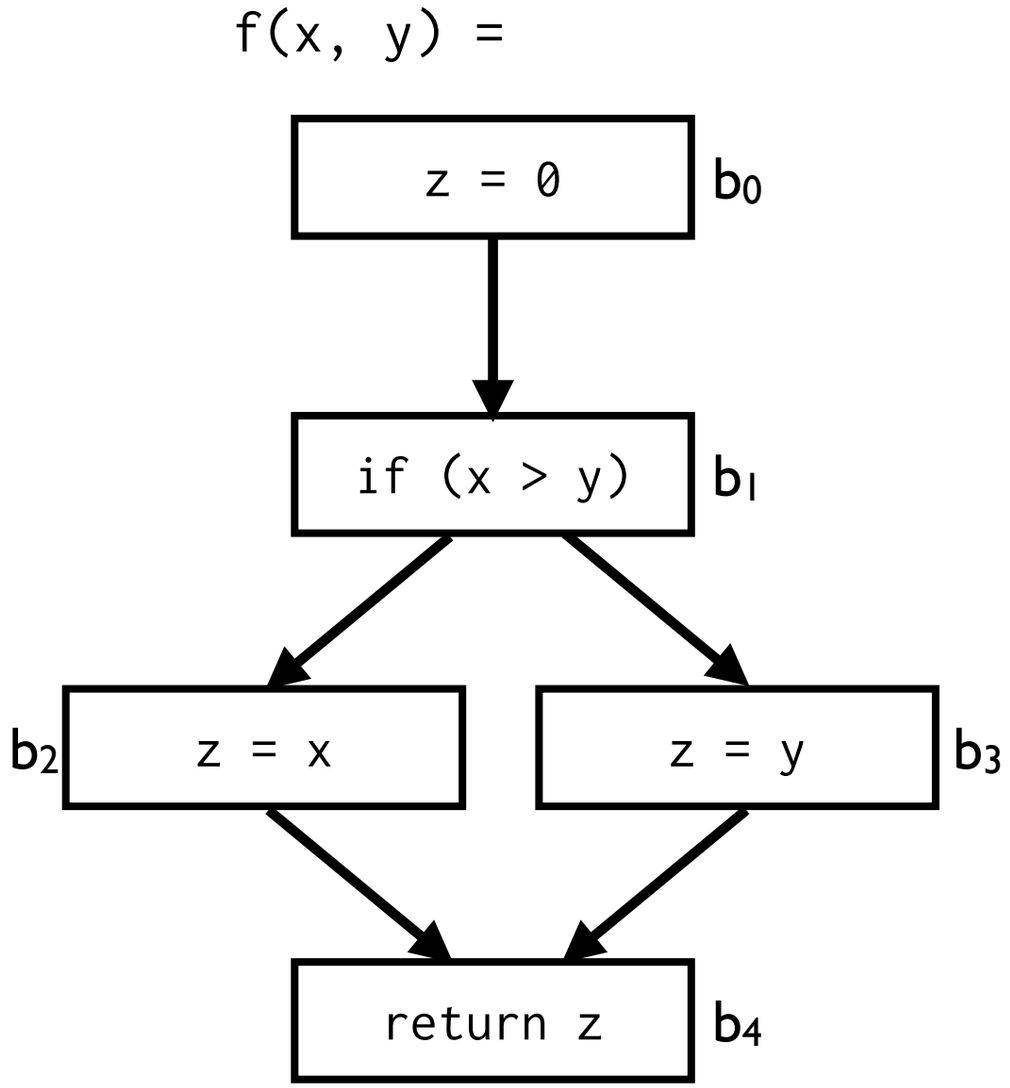

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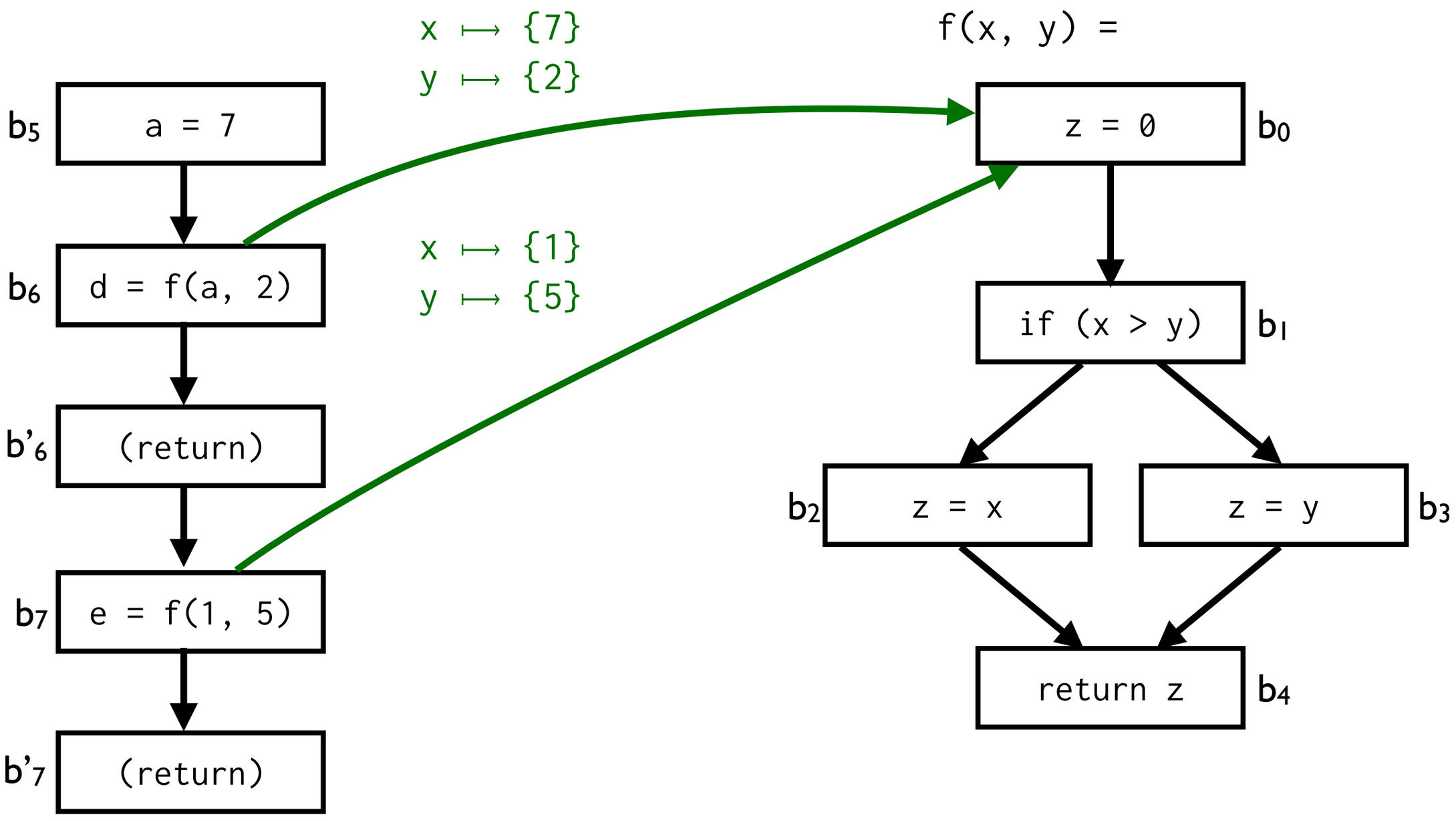

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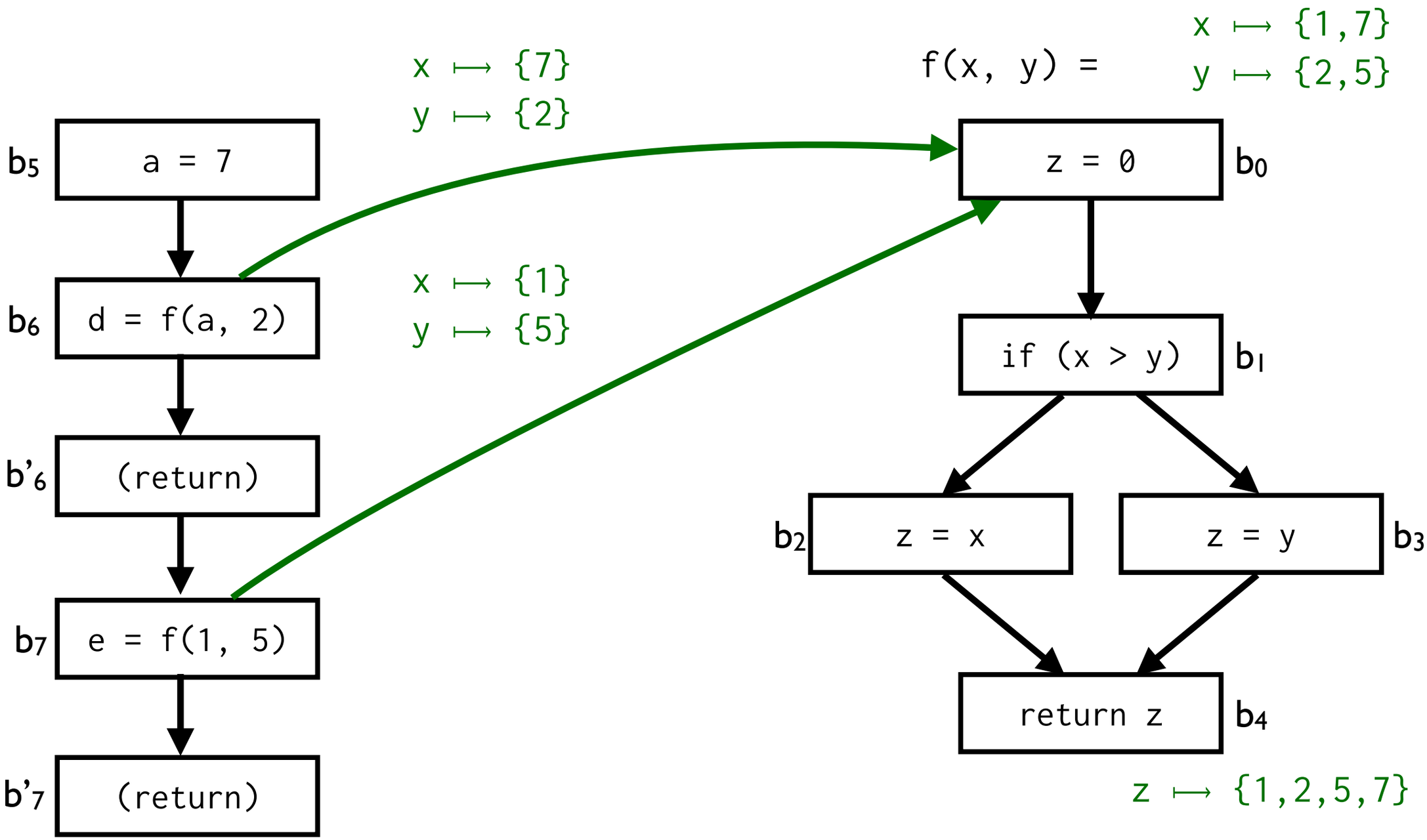

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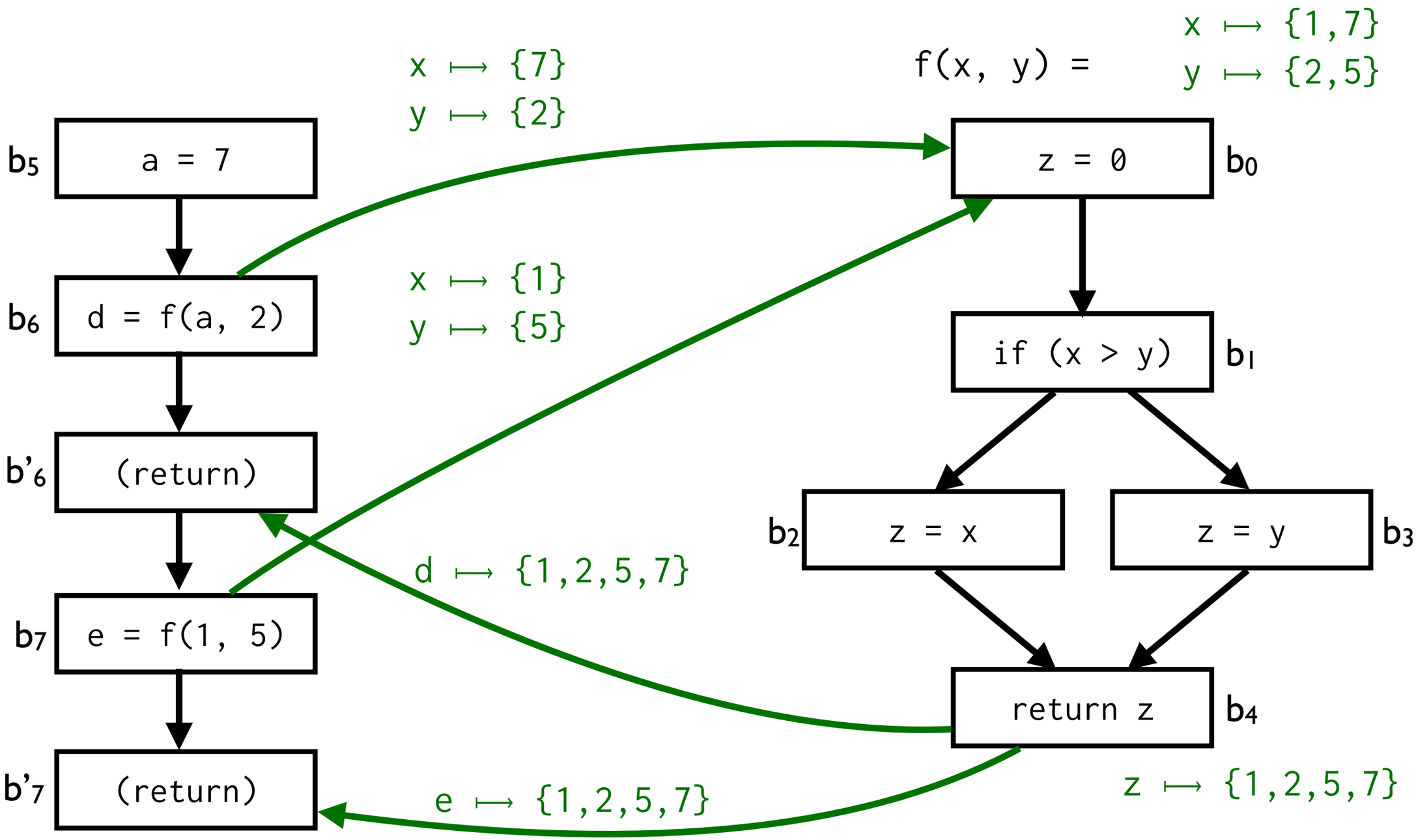


Limitations of Intra-Procedural Analysis

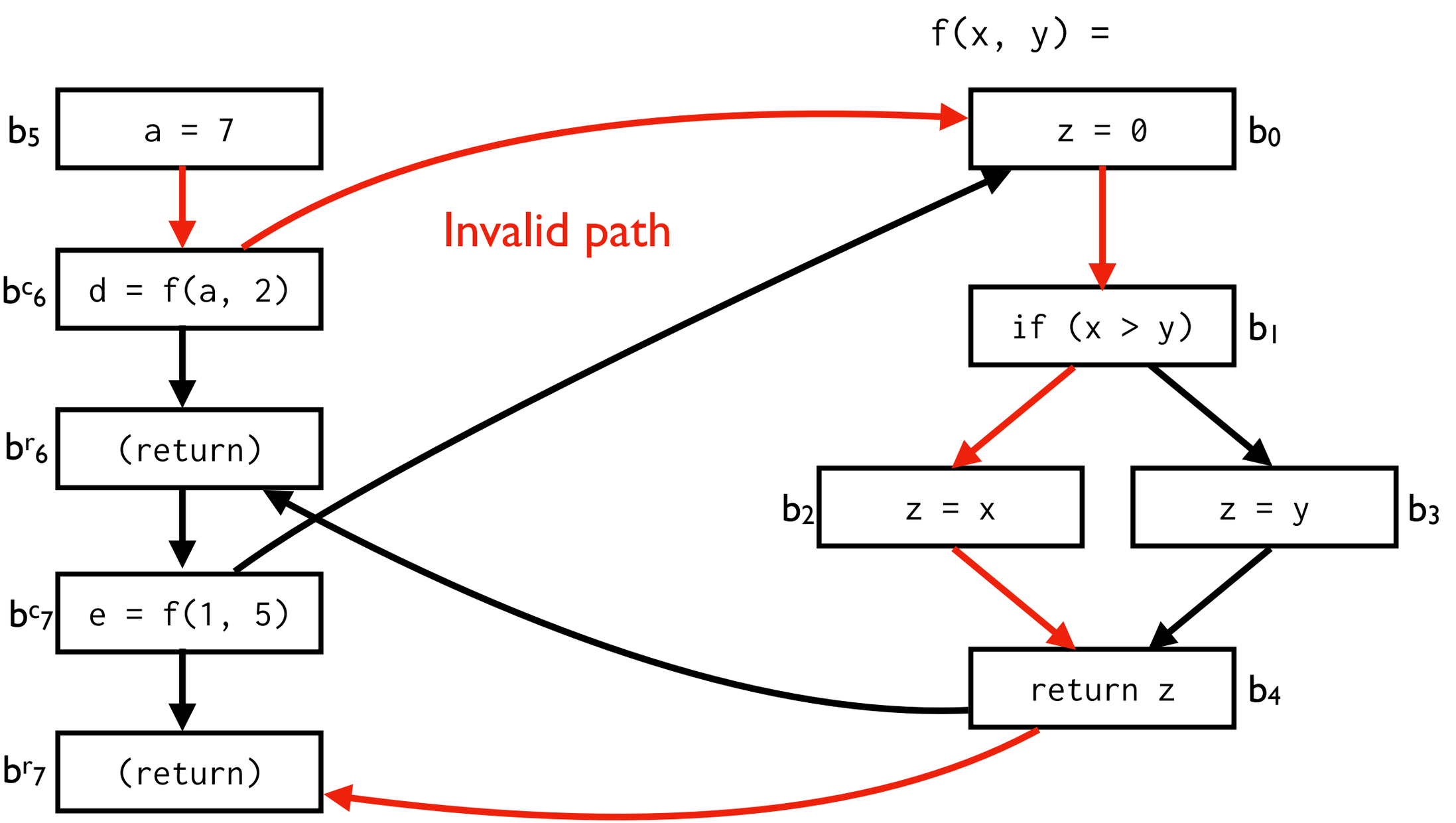
- Split call sites b_x into call (b^c_x) and return (b^r_x) nodes
- Intra-procedural edge:
  - $b^c_x \rightarrow b^c_x$ carries environment/store
- Inter-procedural edge:
  - Caller → subroutine, substitutes parameters (for pass-by-value)
  - Return → Caller, substitutes result (for pass-by-result)

Limitations of Intra-Procedural Analysis

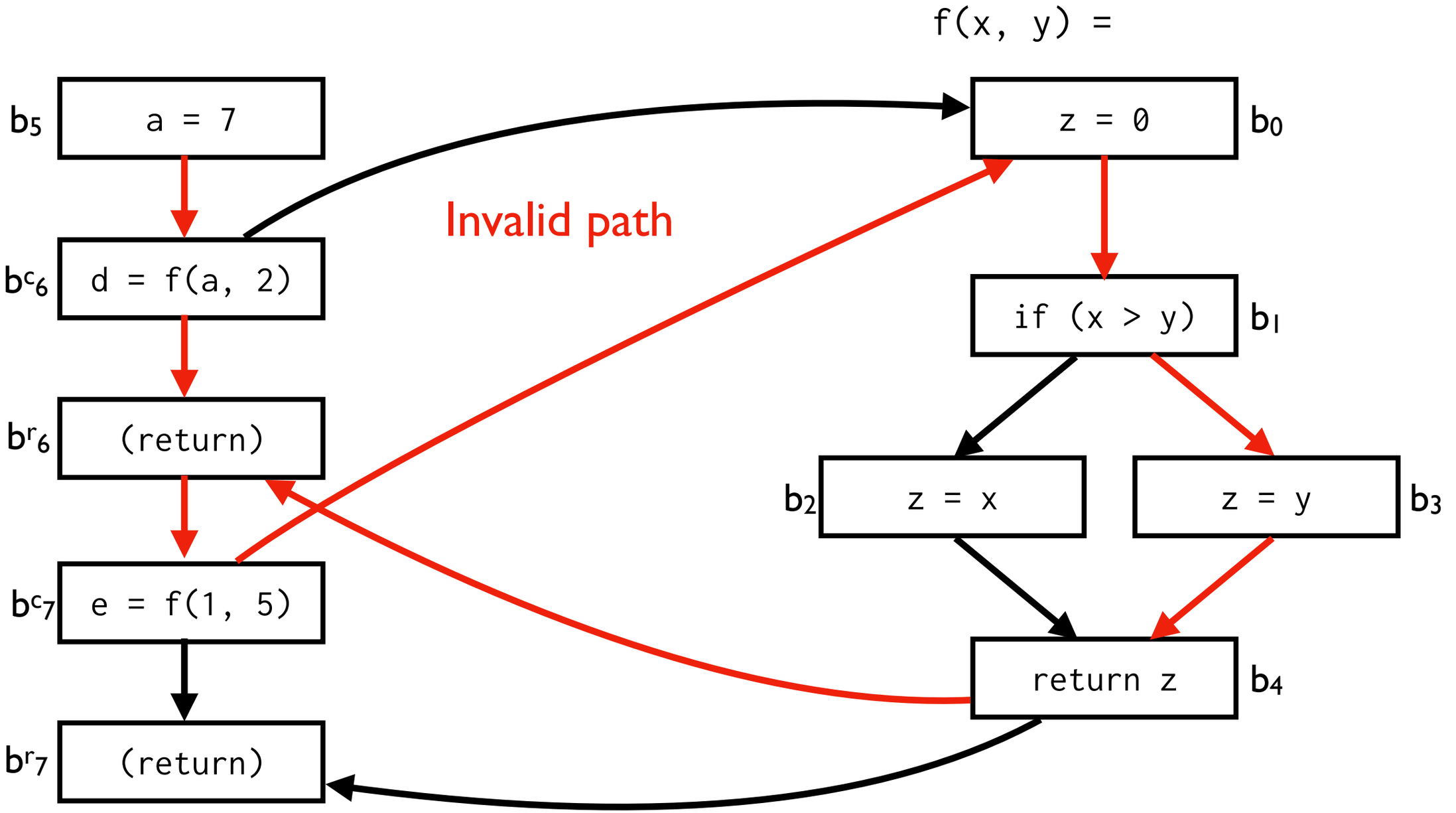
- In this **naive** approach, **all call sites** connected to subroutine entry and subroutine return connected to **all return sites**
- So the analysis is **context-insensitive** (it cannot distinguish between the different call contexts)

Meet over valid paths

Valid Paths

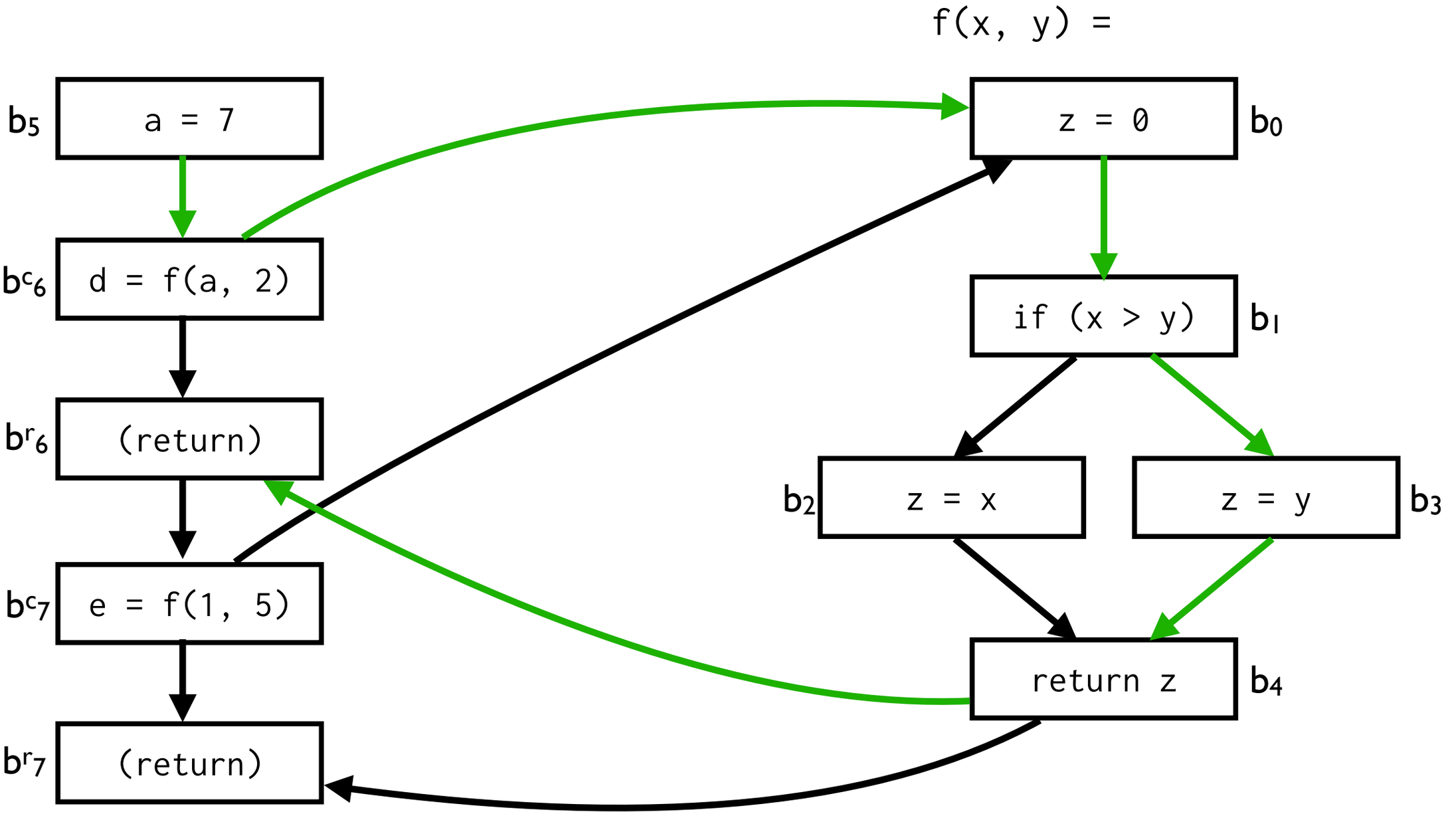

f(x, y) =
b5
a = 7
bc6
d = f(a, 2)
br6
(return)
bc7
e = f(1, 5)
br7
(return)
Invalid path
z = 0
b0
if (x > y)
b1
b2
z = x
z = y
b3
return z
b4

Valid Paths


f(x, y) =
b5
a = 7
z = 0
b0
Invalid path
bc6
d = f(a, 2)
if (x > y)
b1
br6
(return)
b2
z = x
z = y
b3
bc7
e = f(1, 5)
return z
b4
br7
(return)

Valid Paths

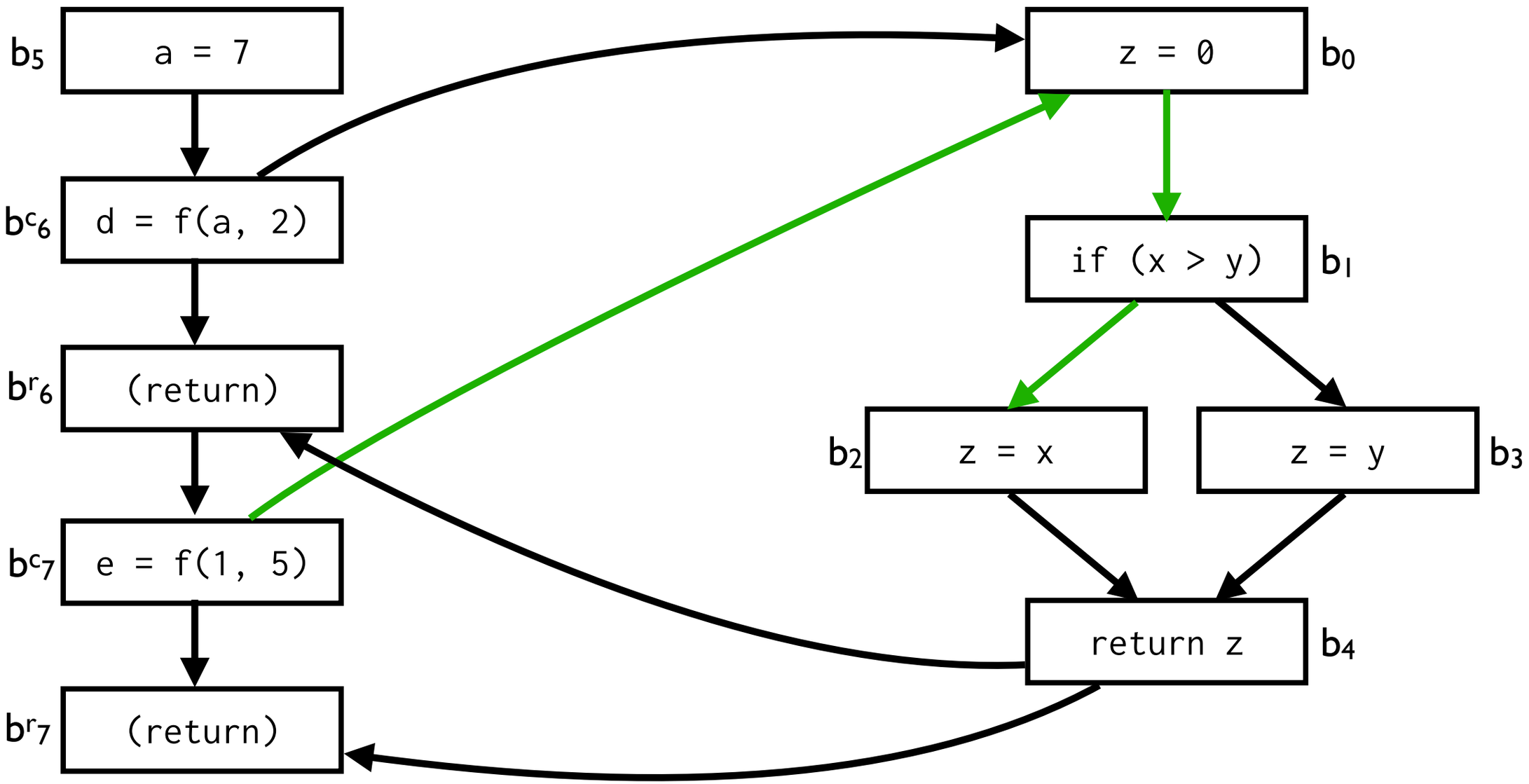
[b_5, b^c_6, b_0, b_1, b_3, b_4, b^r_6,]



Valid Paths

$[b_5, b^c_6, b_0, b_1, b_3, b_4, b^r_6]$

$[b^c_7, b_0, b_1, b_2]$



Valid Paths

- vpath_b: Set of all valid paths that reach b

$$\text{Path} ::= \langle P \rangle^* \langle M \rangle$$

$$P \quad ::= b_x \mid b^c_y$$

$$M \quad ::= b^c_x \langle M \rangle b^r_x \mid b_y \langle M \rangle \mid \langle M \rangle b_y \mid \varepsilon$$

- M productions: $b^c_x \langle M \rangle b^r_x$:

  call and return must **match up**

- P productions (prefix): b^c_y:

  can have **calls that we never return from**

Meet over Valid Paths (MVP)

- Intra-procedural analysis aims for MOP
  - Undecidable
- Now we aim for its **context-sensitive** version:

$$\mathbf{out}_{bi} = \bigsqcup_{[p0, \ldots pk] \in \mathbf{vpath}_{bi}} \mathbf{trans}_{p0} \circ \ldots \circ \mathbf{trans}_{pk} \circ \mathbf{trans}_{bi} (\bot)$$

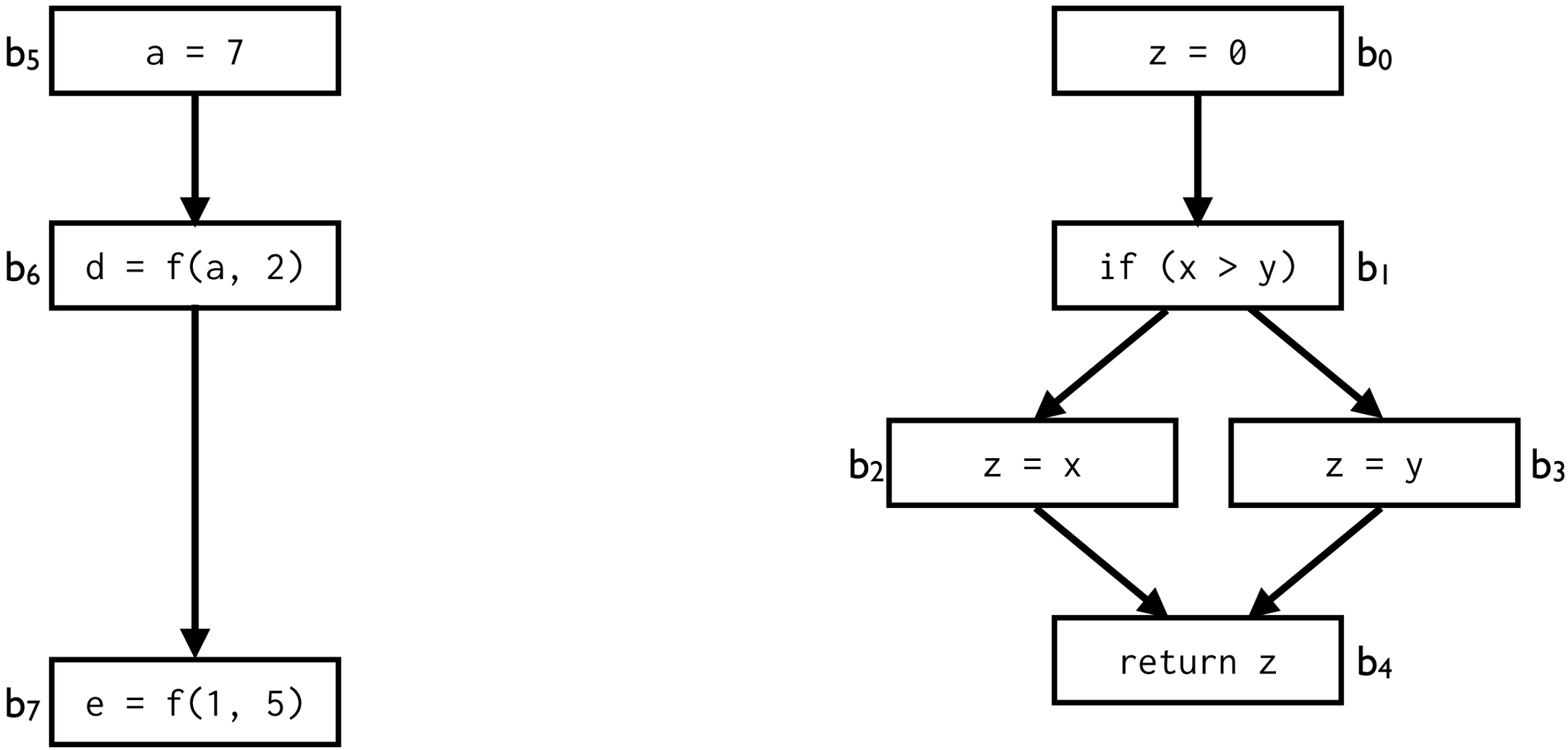
  - Undecidable, like MOP

Context Sensitivity

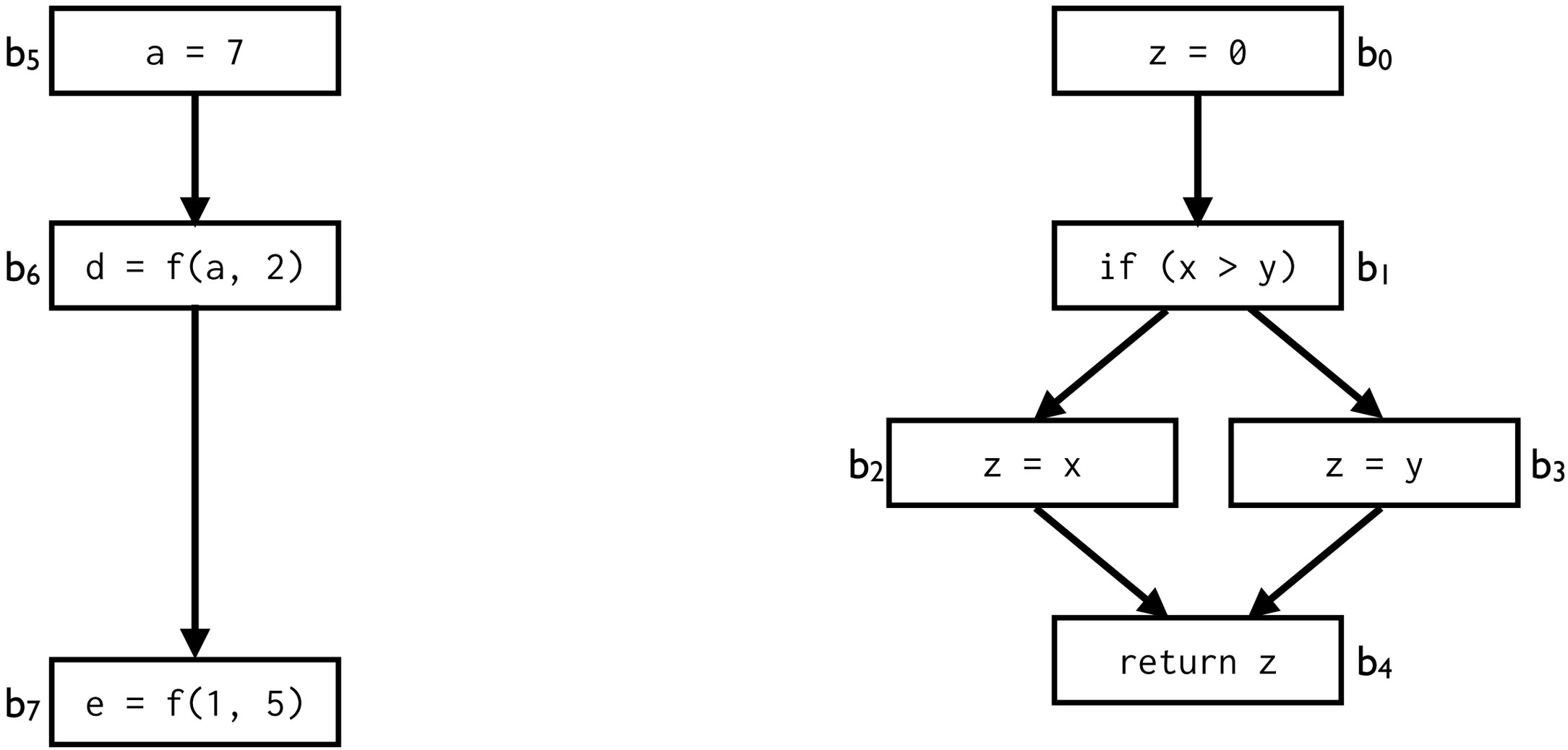
Context-Sensitivity

- Goal: distinguish among several context calls.
- Techniques:
  - Cloning
  - Inlining
  - Call Stacks/Call Strings
  - Procedure Summaries

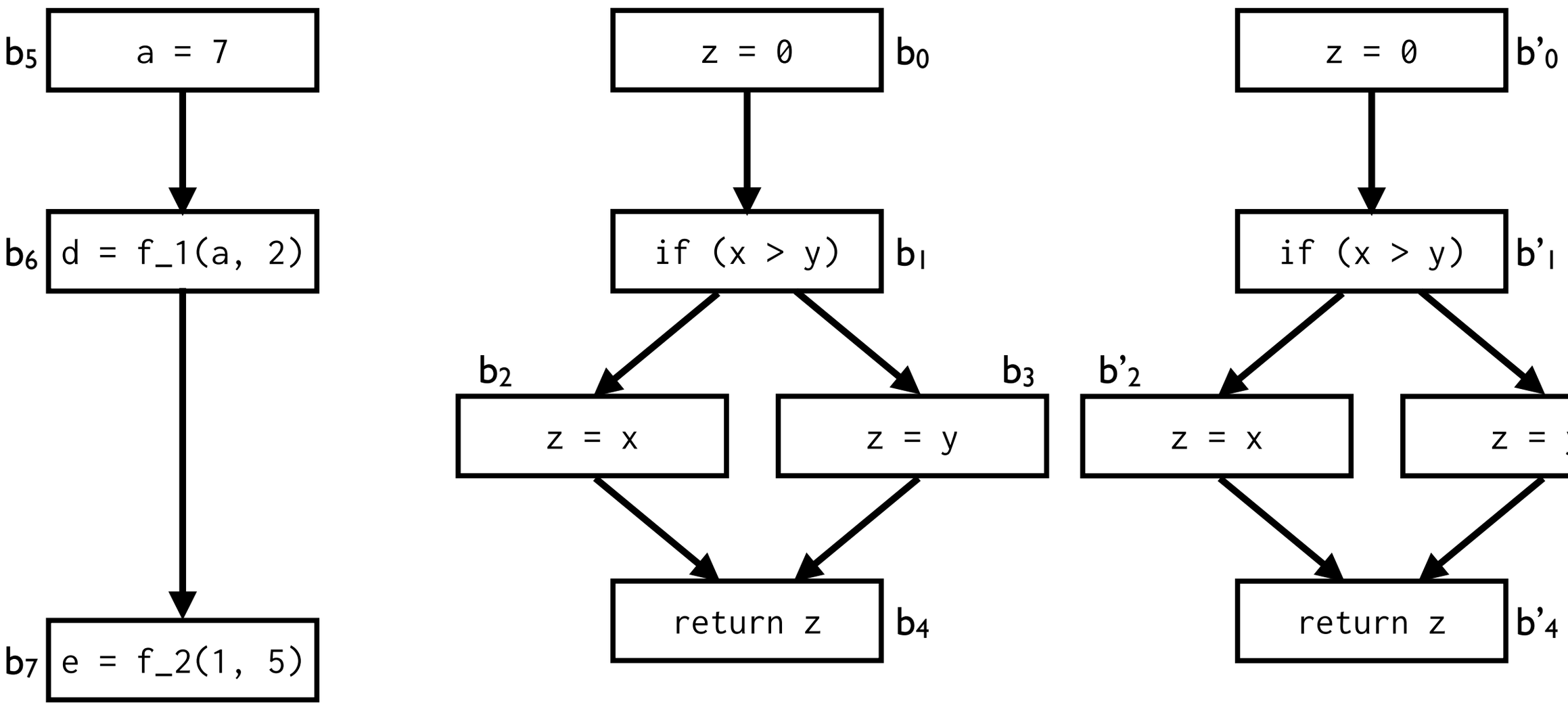
Context Sensitivity: Cloning



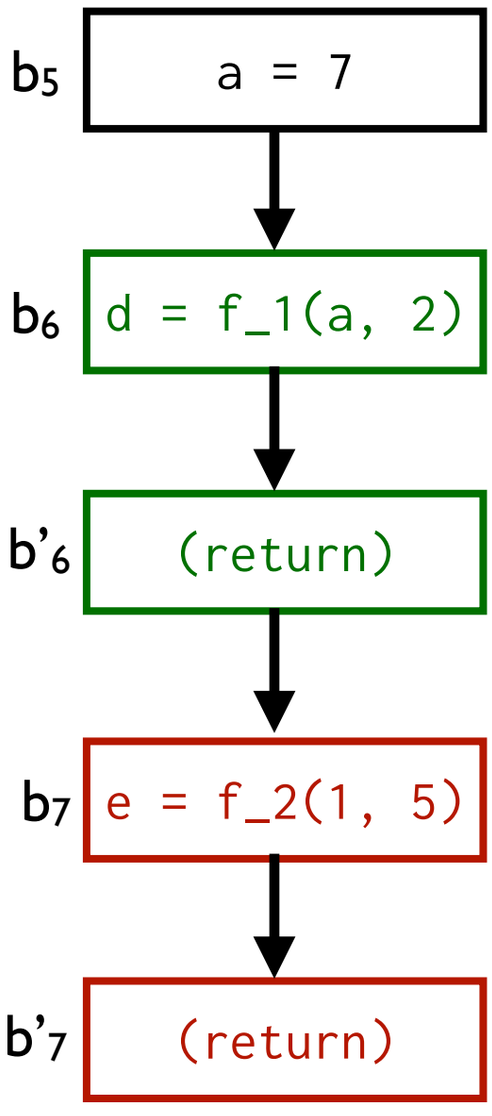
Context Sensitivity: Cloning

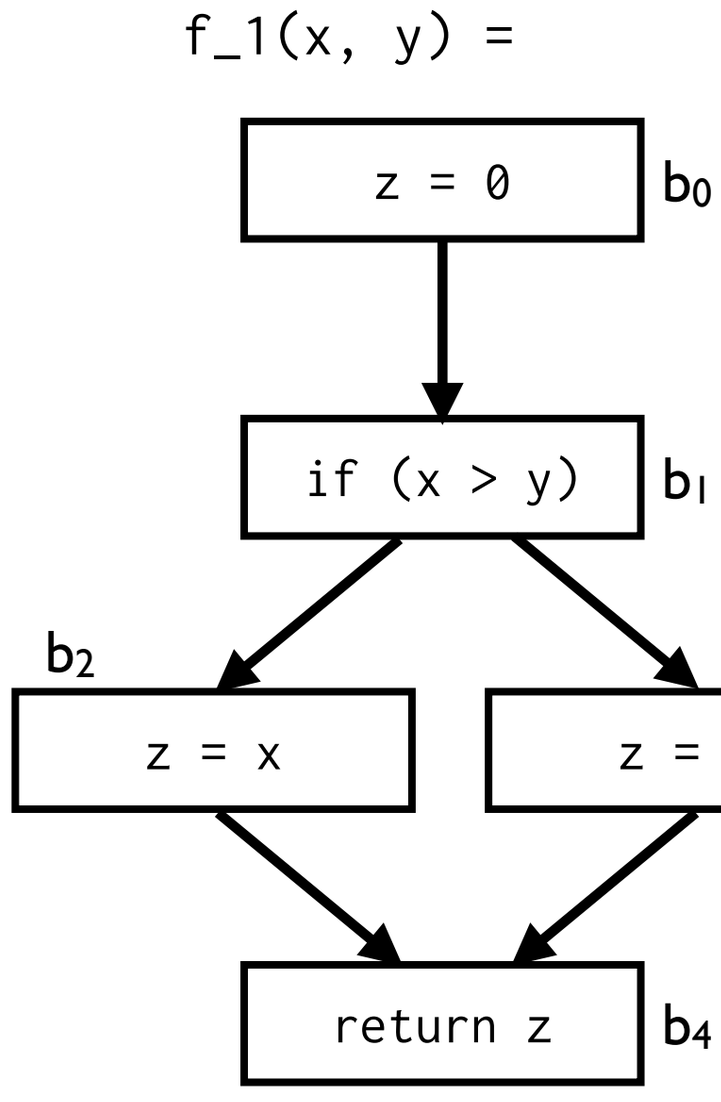


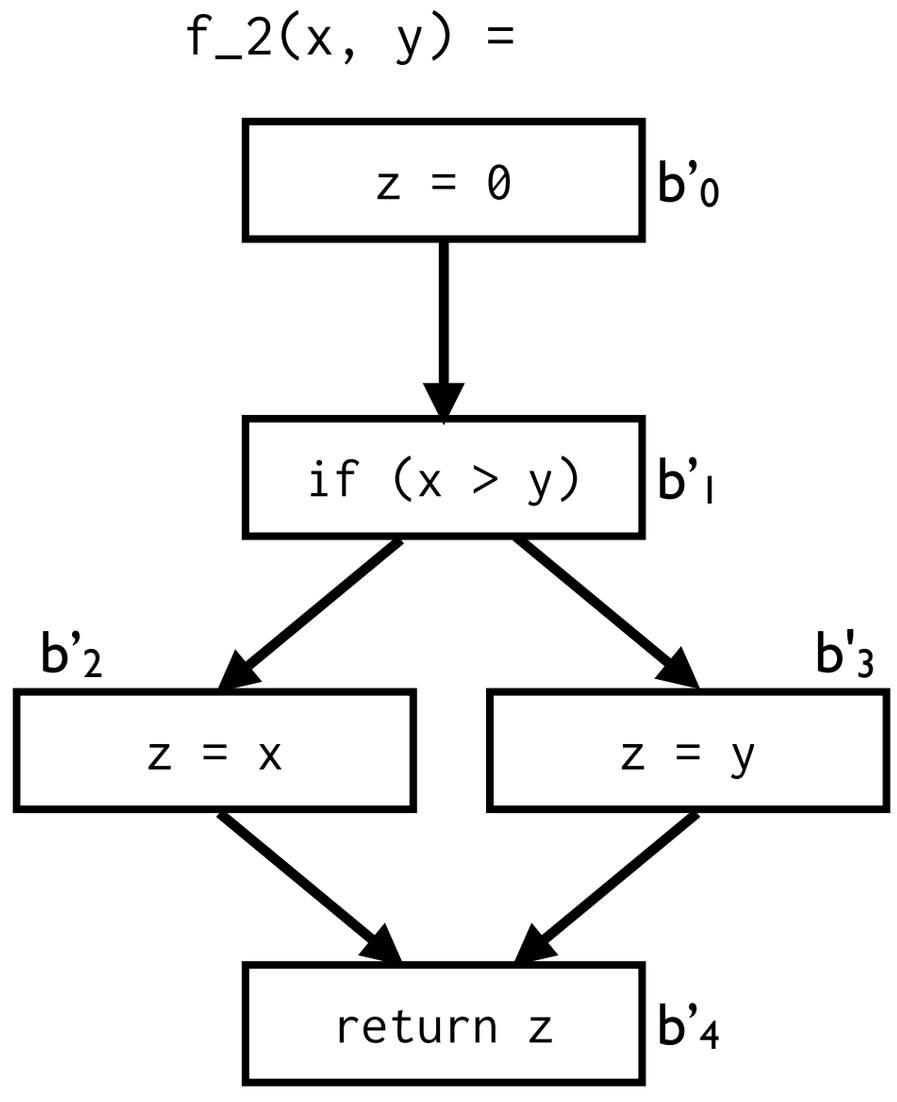
Context Sensitivity: Cloning



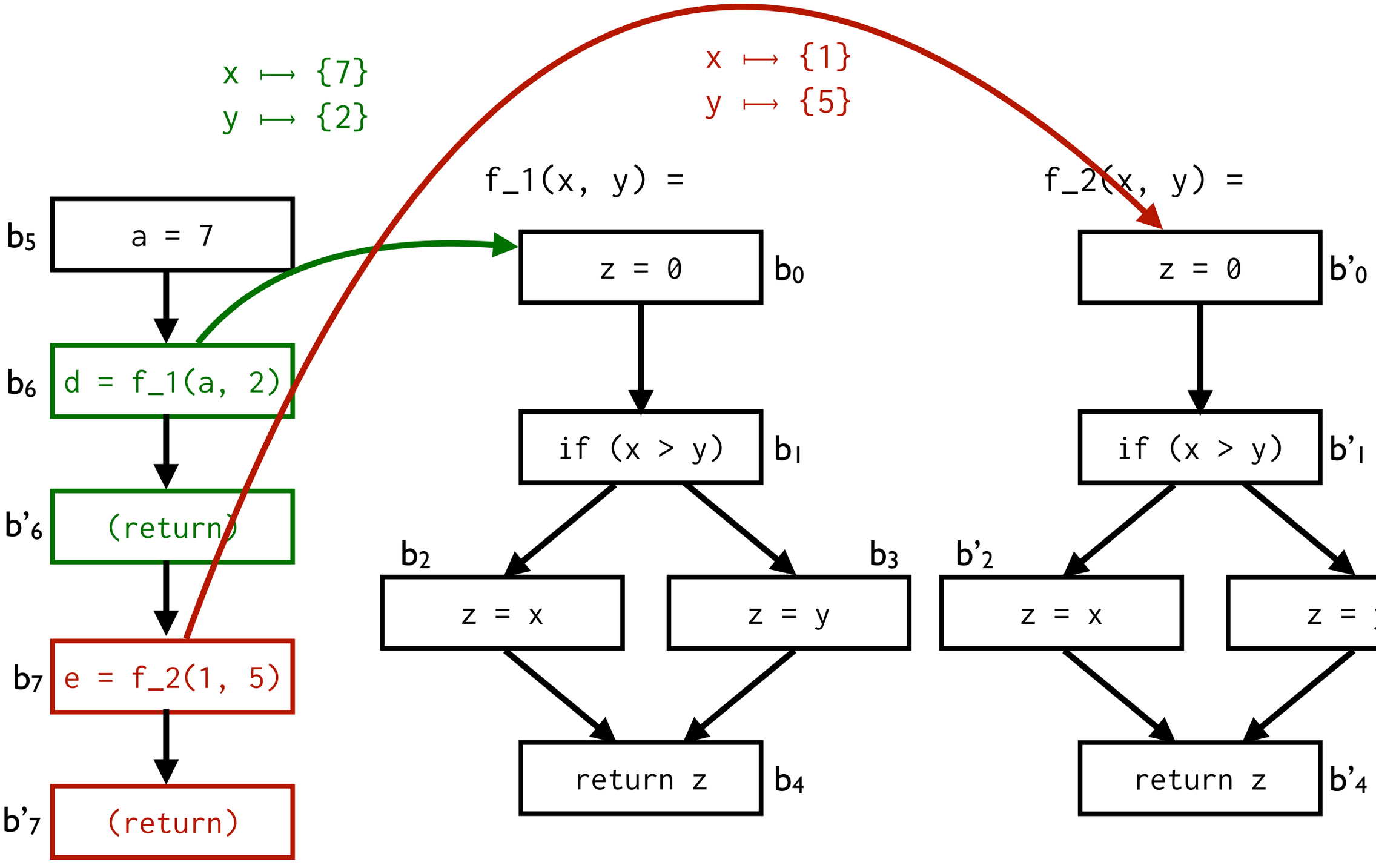
Context Sensitivity: Cloning



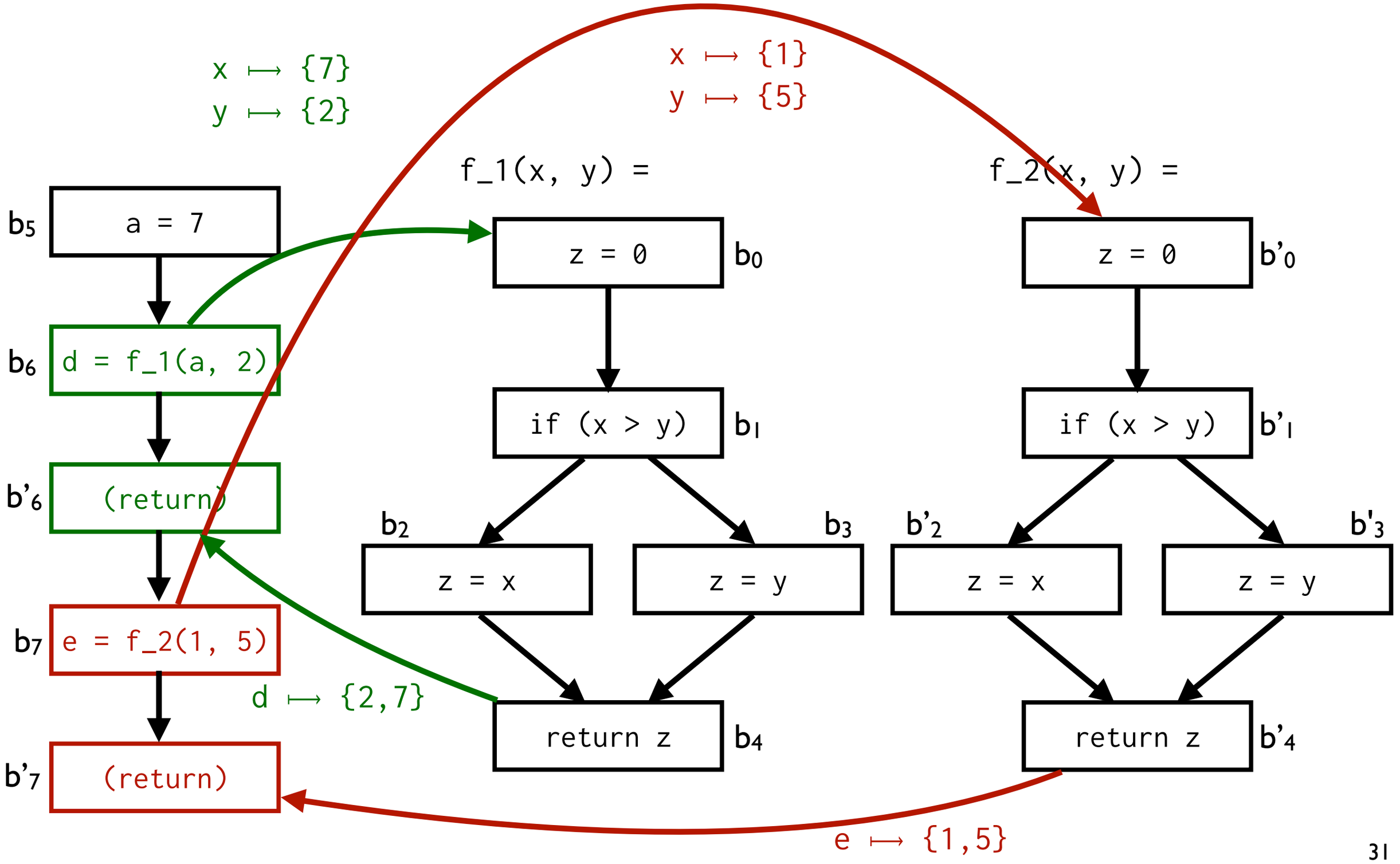




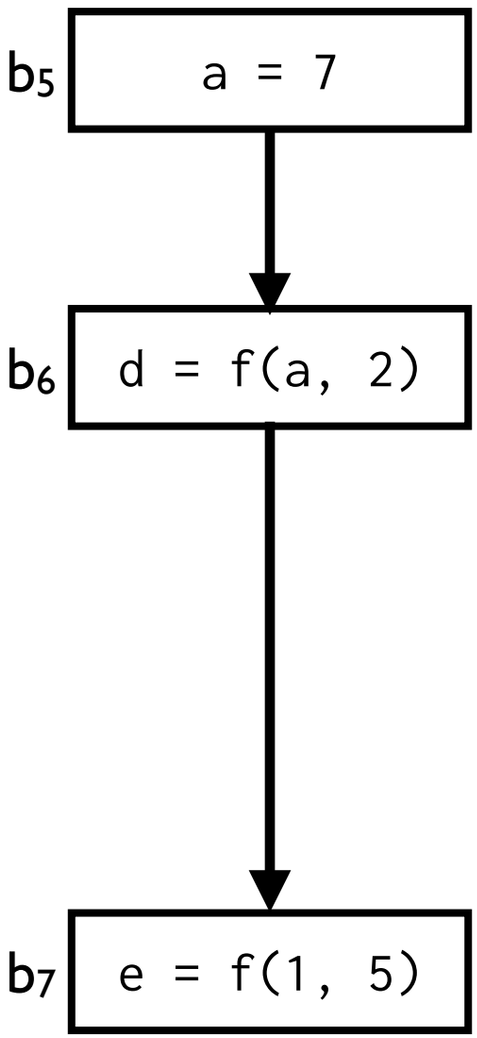
Context Sensitivity: Cloning

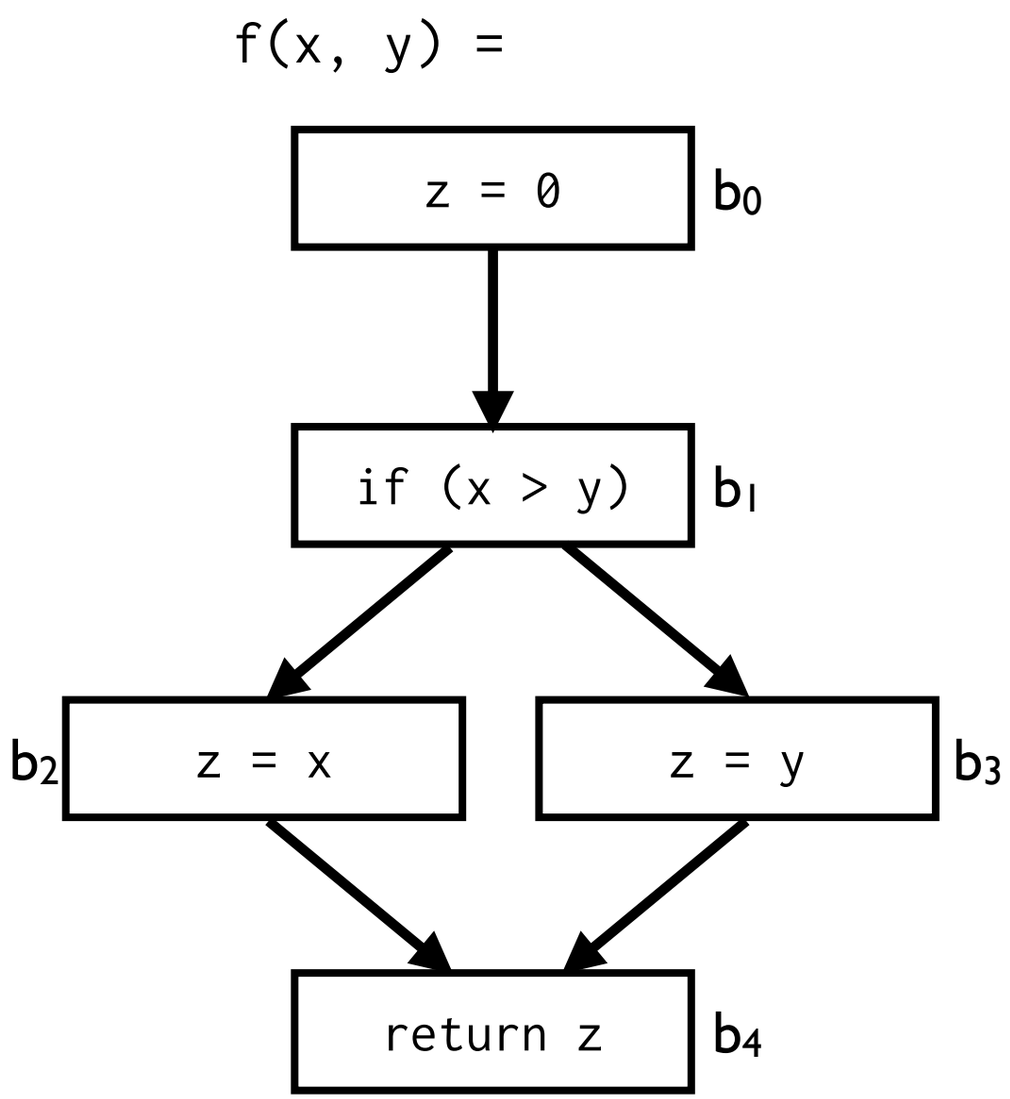


Context Sensitivity: Cloning

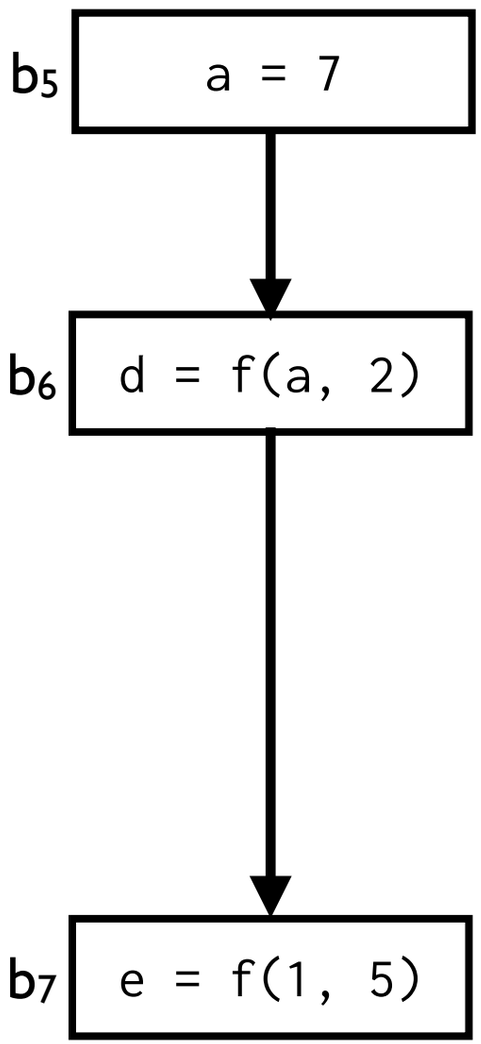


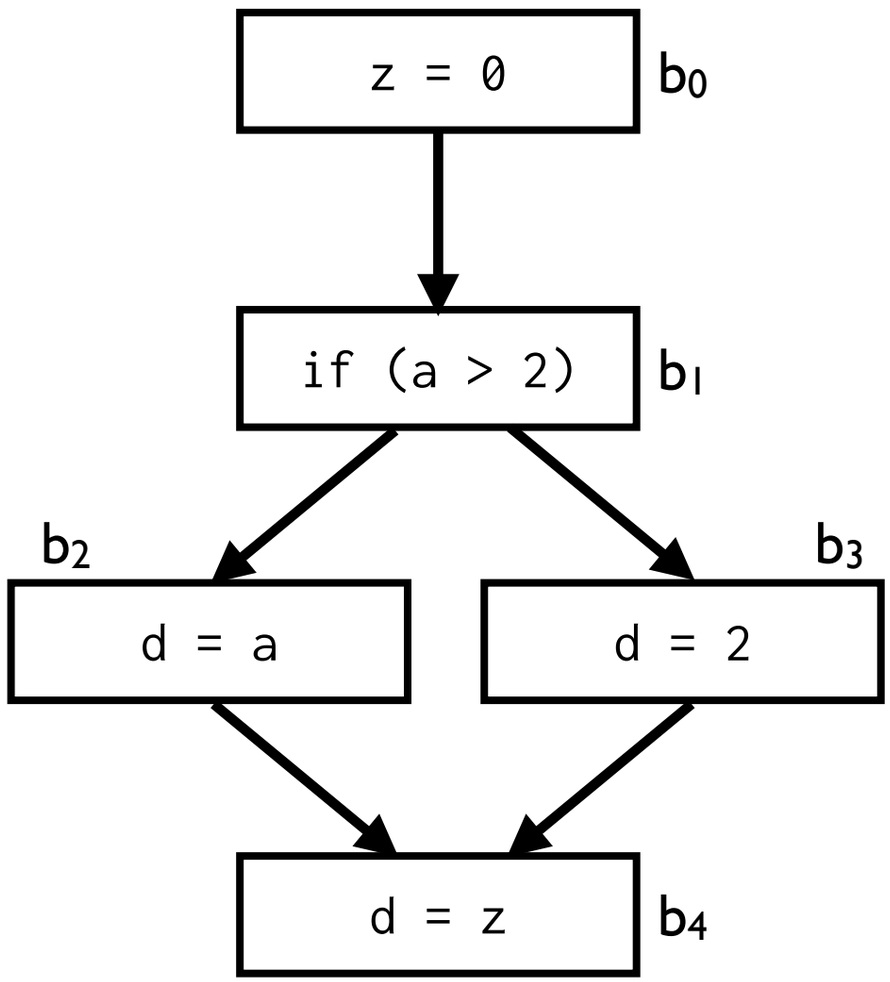
Context Sensitivity: Inlining

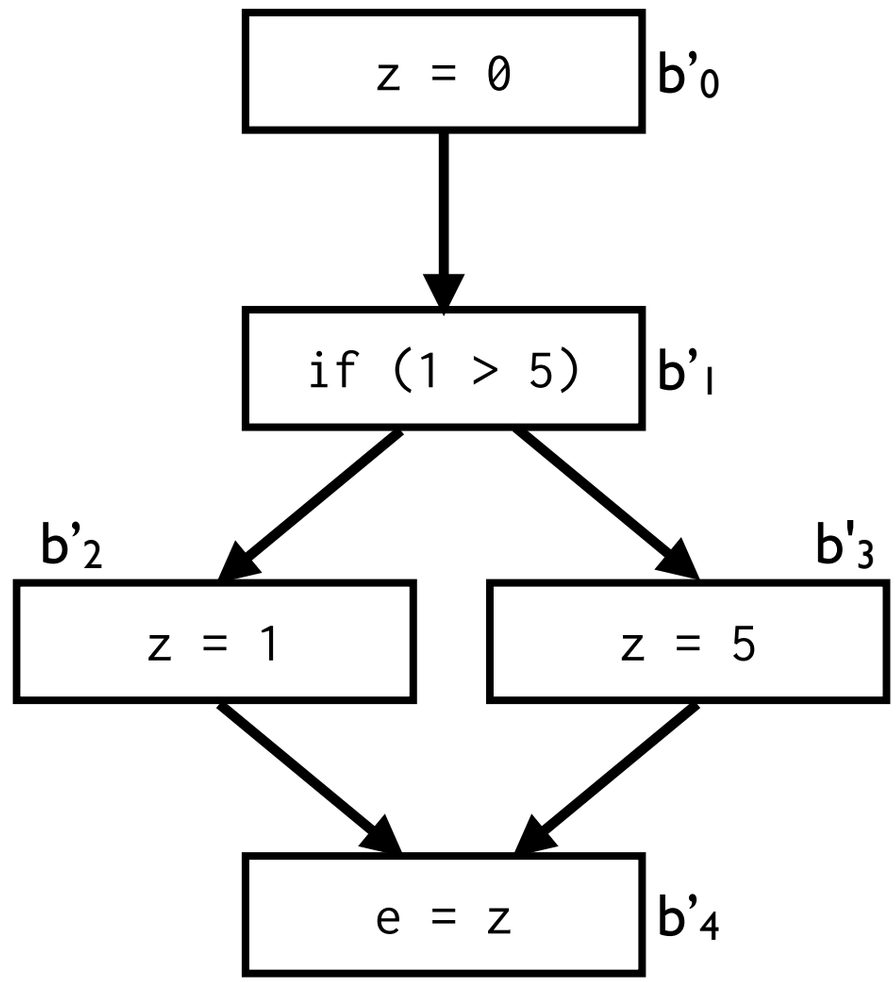




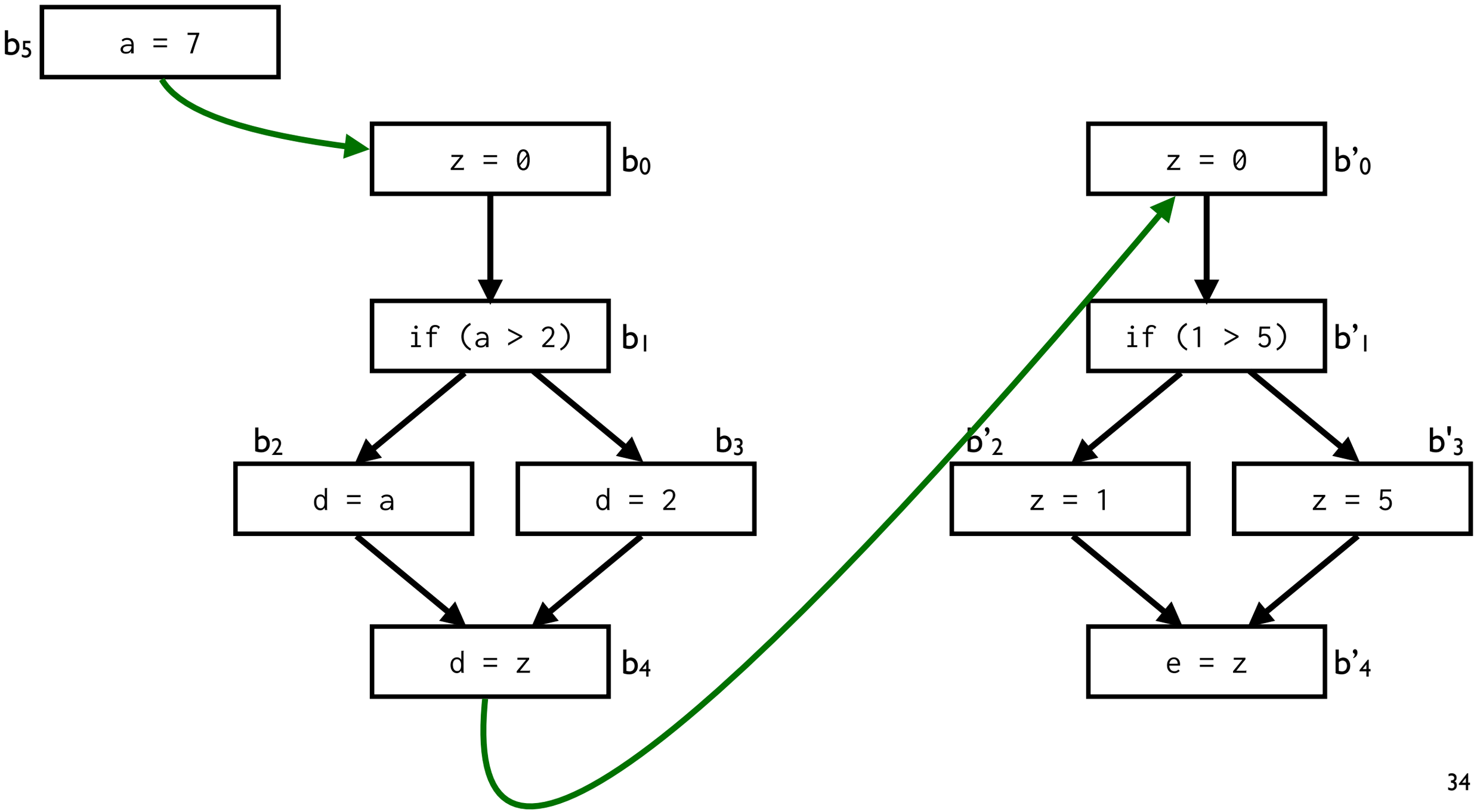
Context Sensitivity: Inlining



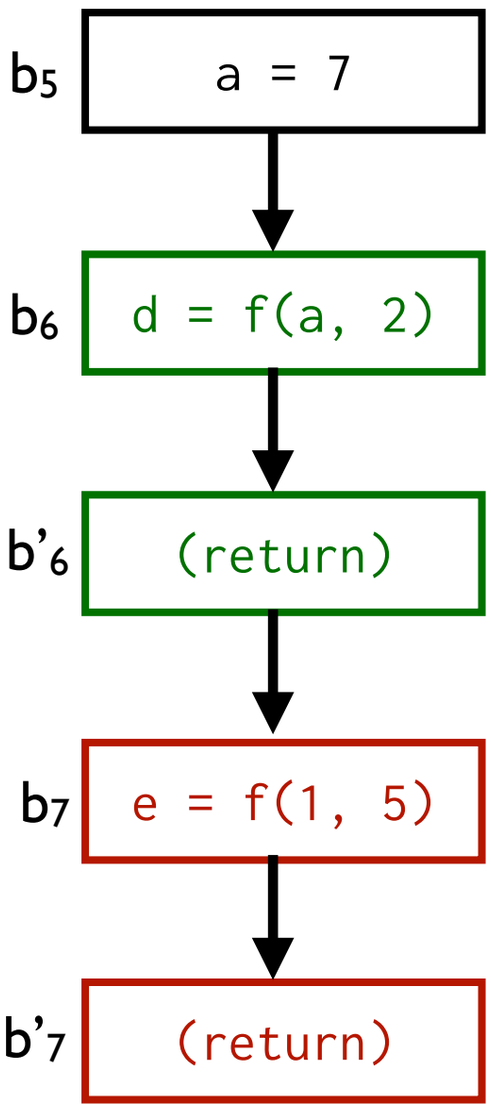


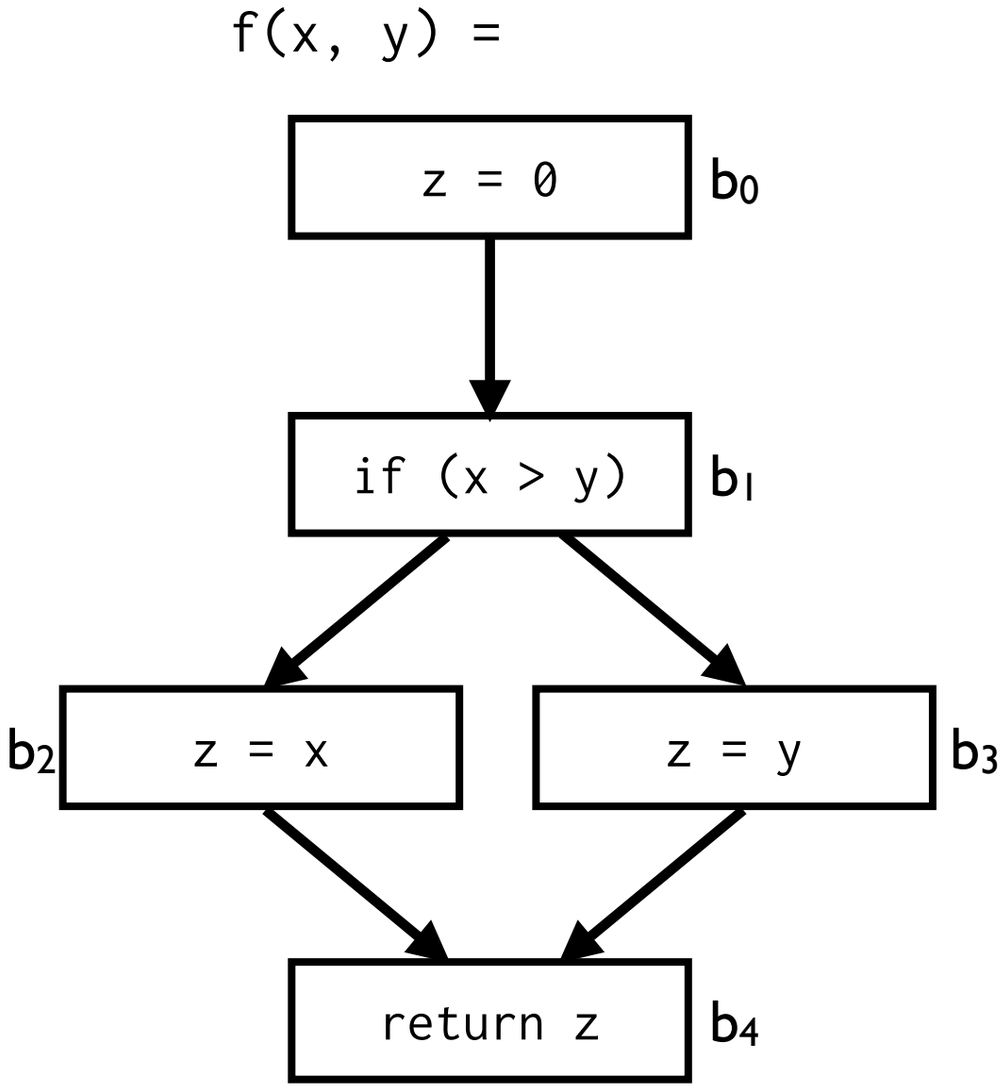


Context Sensitivity: Inlining



Context Sensitivity: Call Strings





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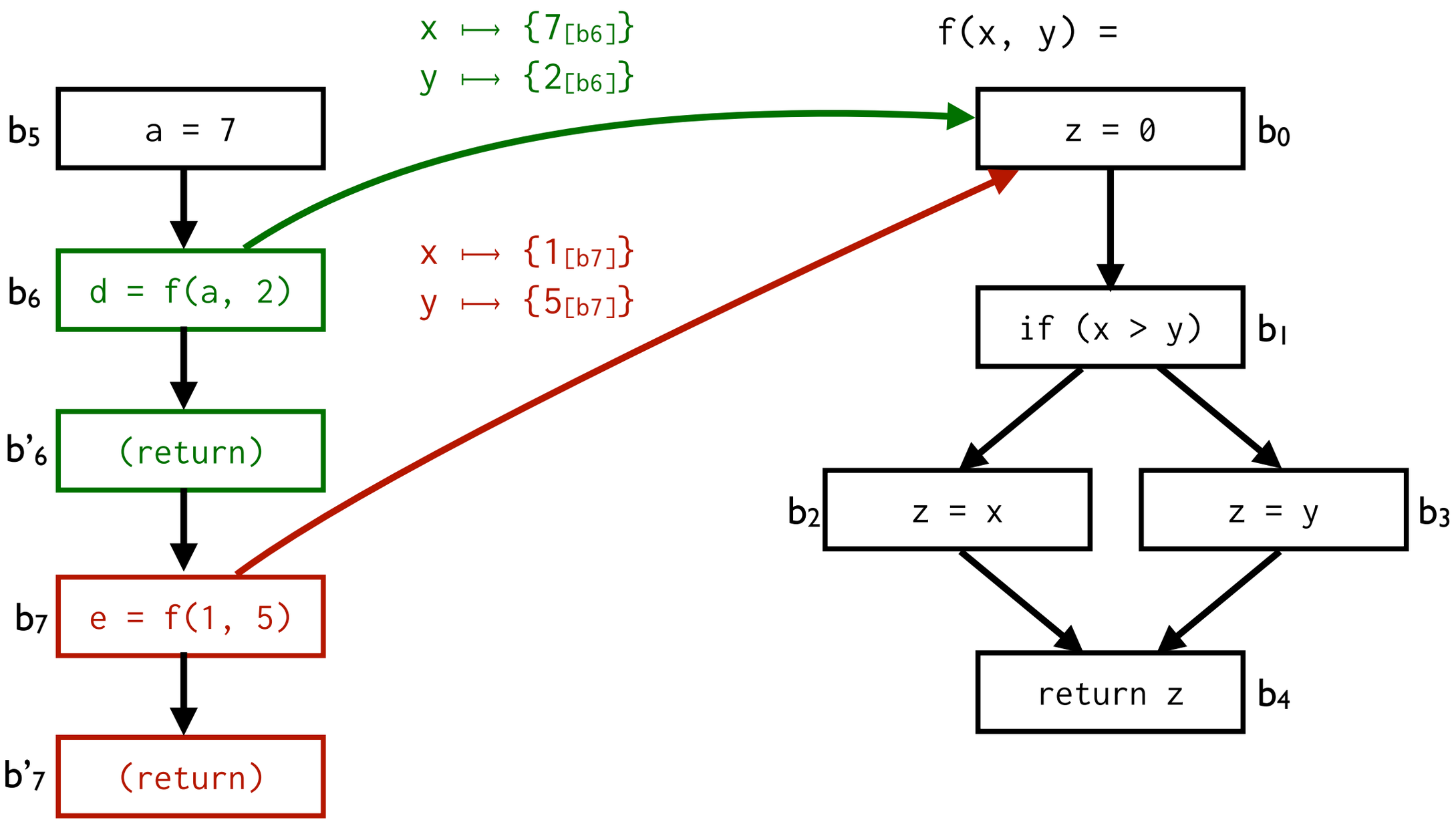

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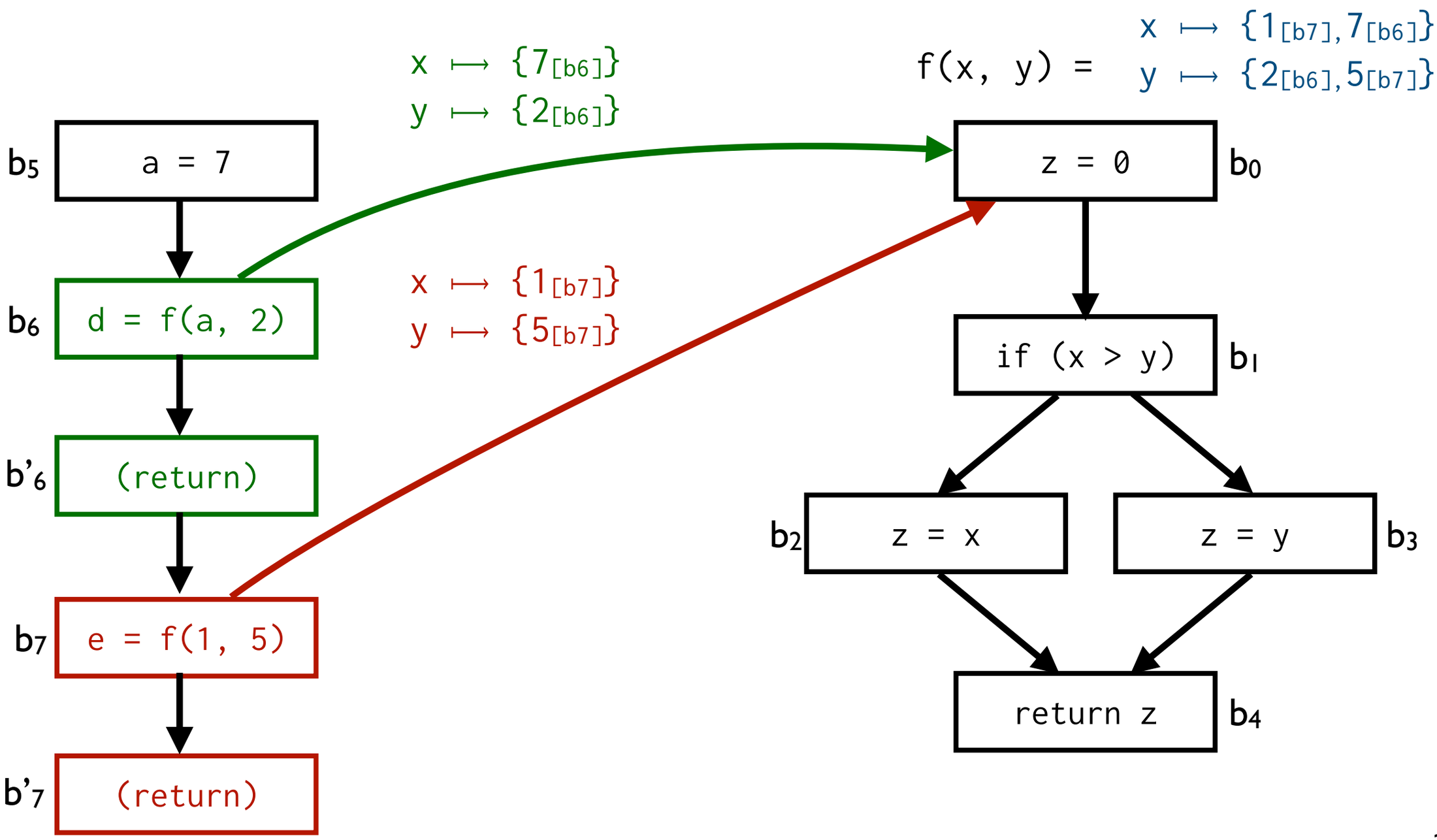

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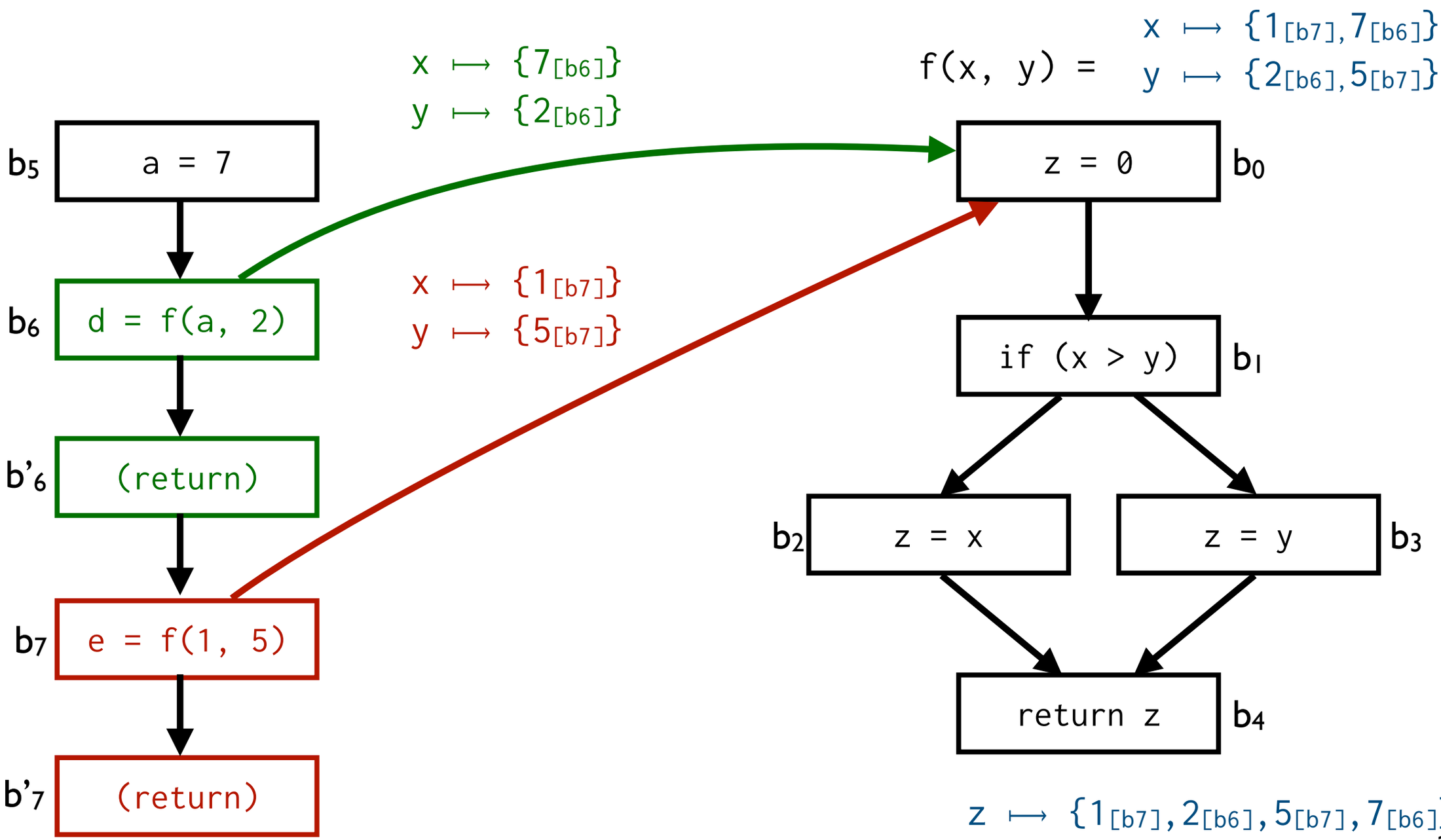

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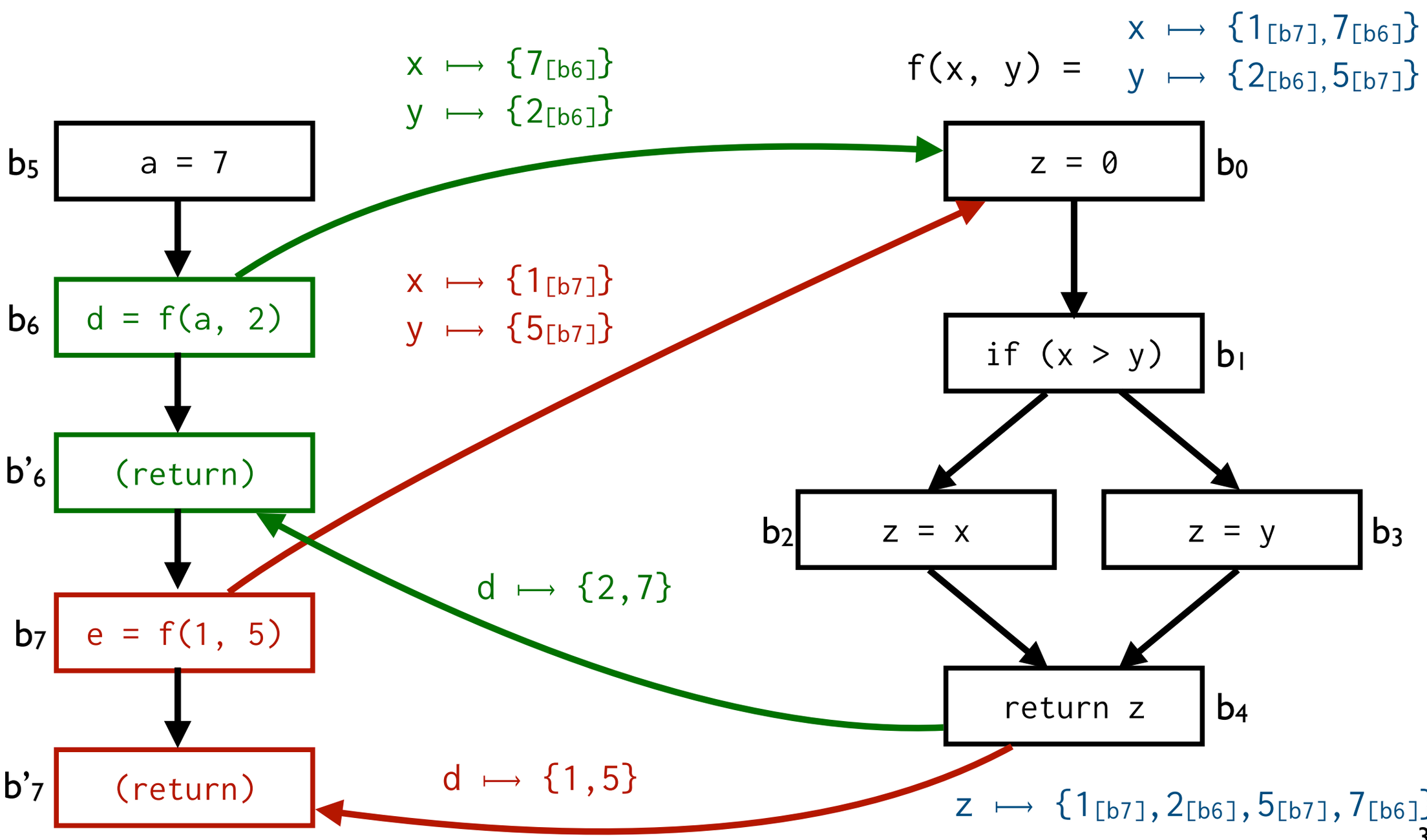


Context Sensitivity: Call Strings

- We used call sites to make context explicit
- Generalization:
- Call Strings support deeper nesting
- Examples: [b0, b6], [b1, b6]
- Must bound length of call strings to ensure termination

```
int g(y) {
  return y;
}

int f(x) {
  return g(x)
       + g(5);
}

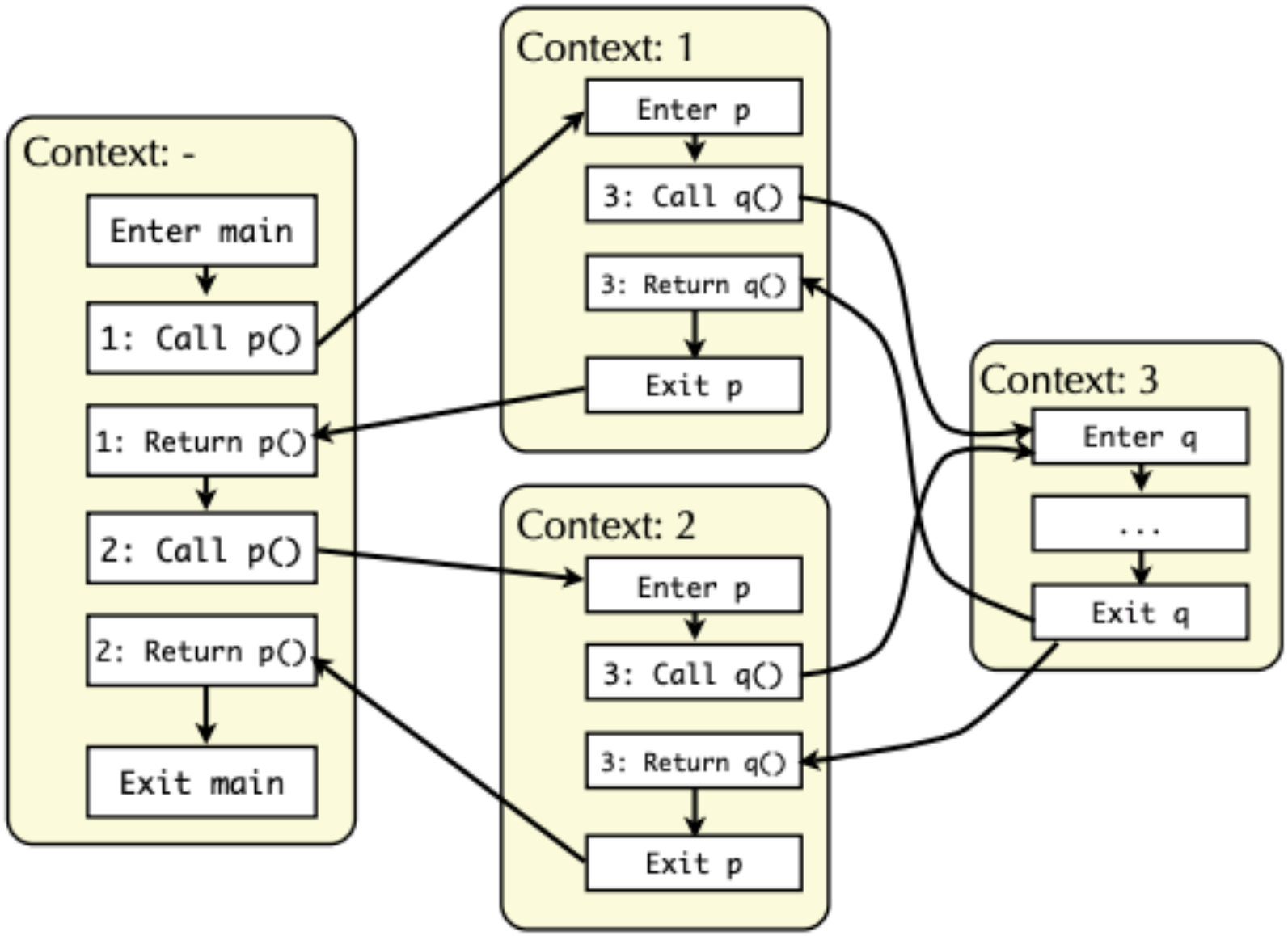
…

f(1);
f(2);
```

Context Sensitivity: Call Strings

```
main() {
  1: p();
  2: p();
}

p() {
  3: q();
}
q() {
  …
}
```

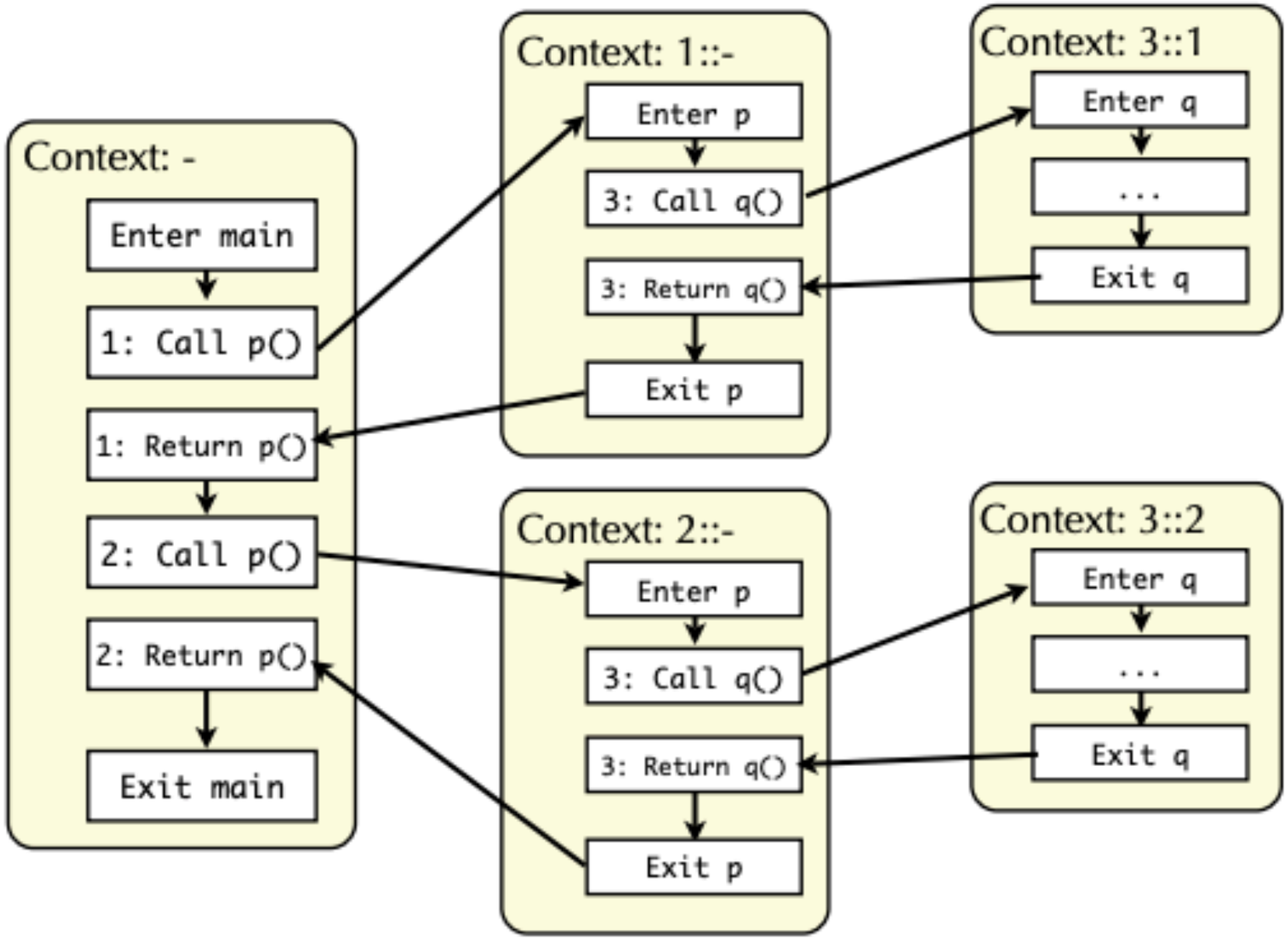


Call strings of size 1

Context Sensitivity: Call Strings

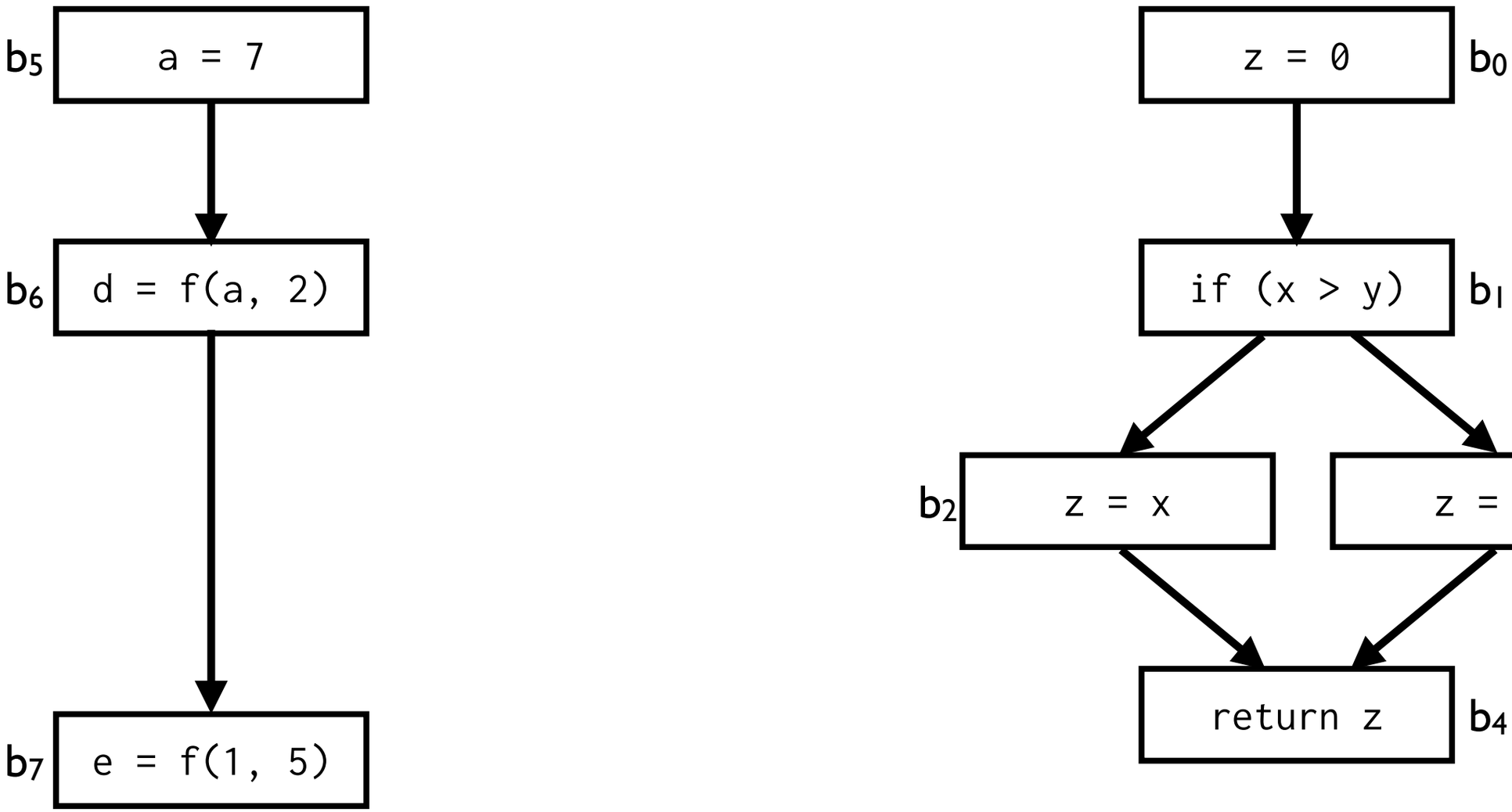
```
main() {
  1: p();
  2: p();
}

p() {
  3: q();
}
q() {
  …
}
```

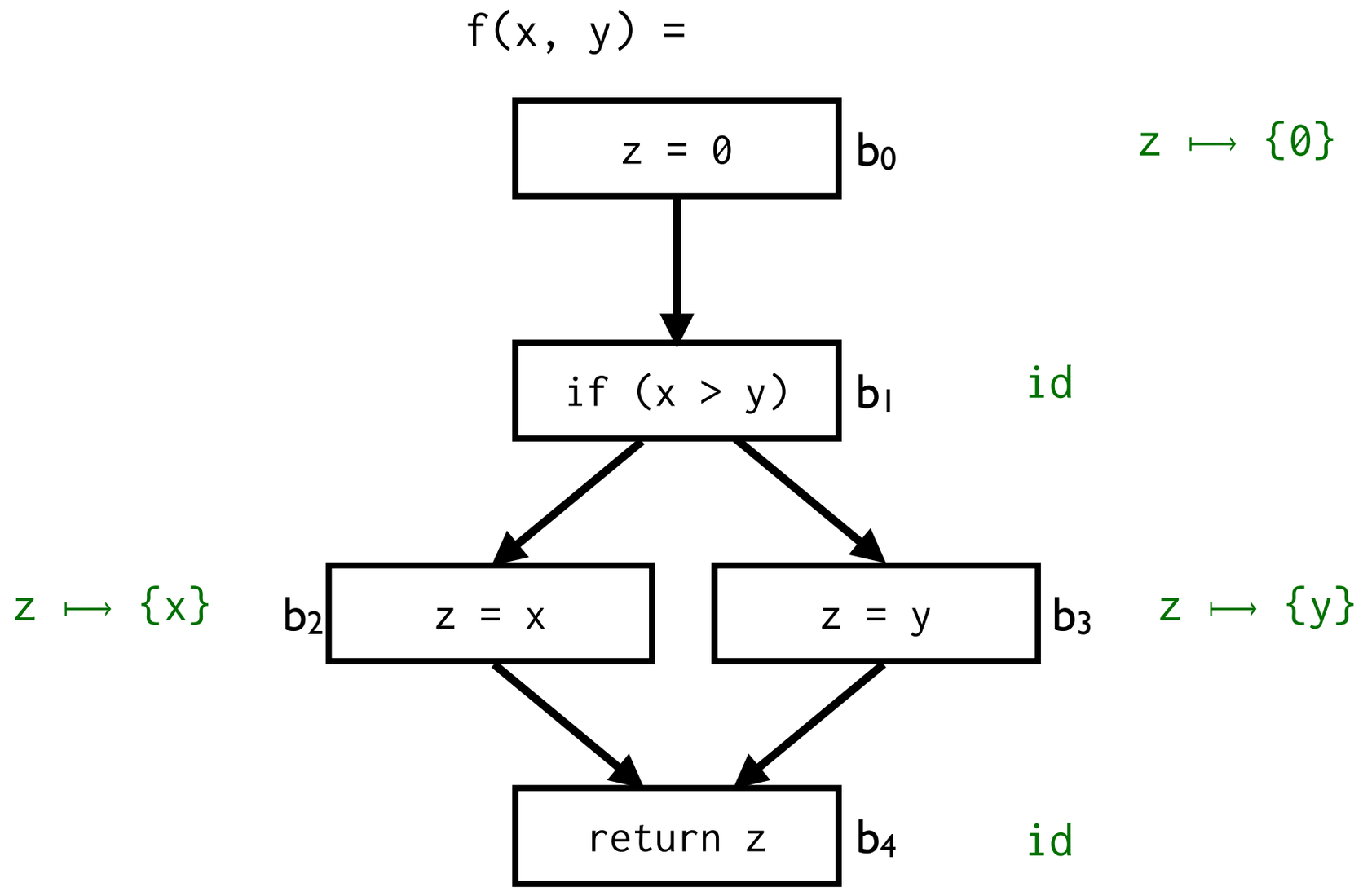


Call strings of size 2

Context Sensitivity: Procedure Summaries

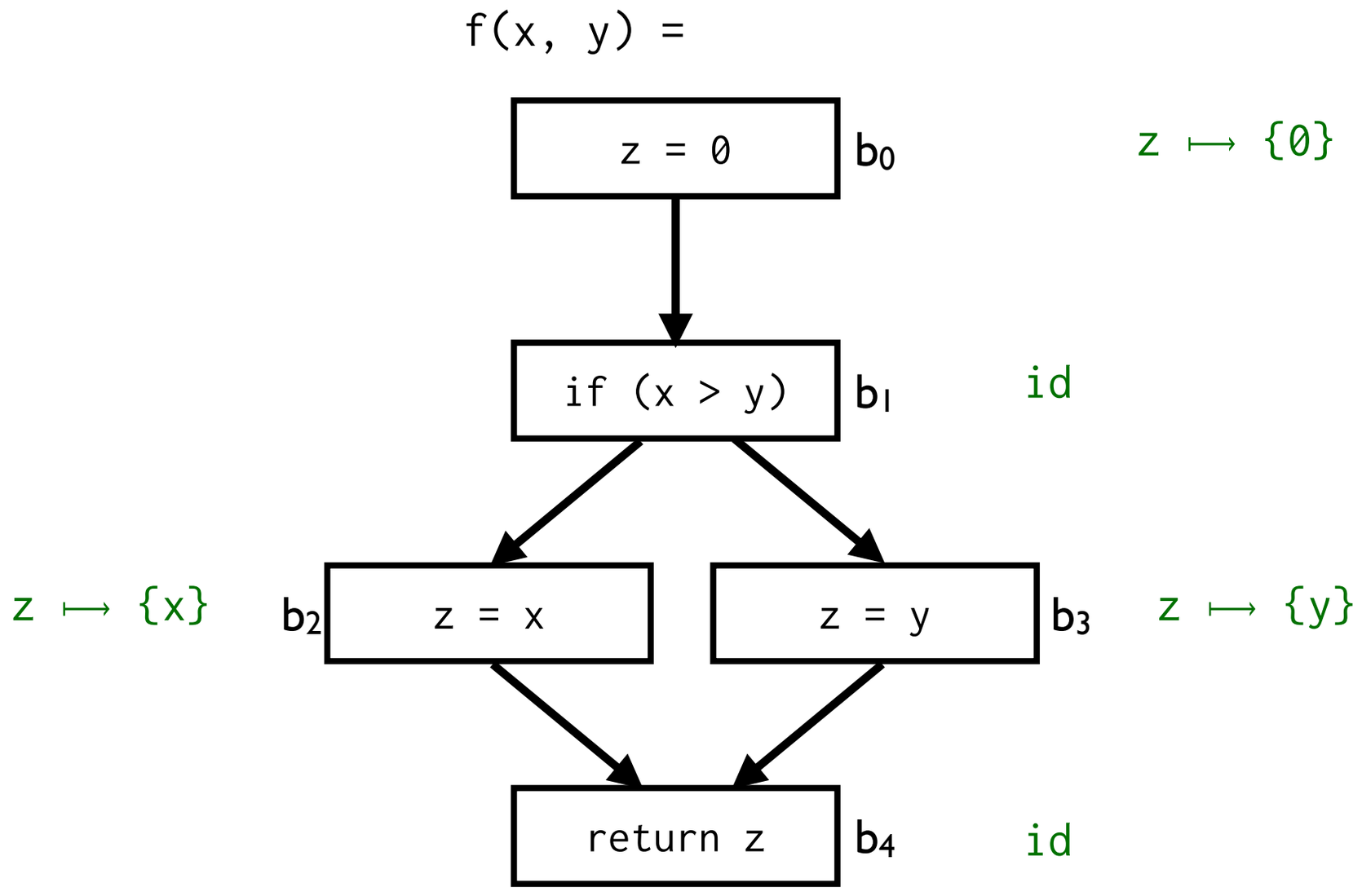


Composed Transfer Function



$trans_{b0} \circ trans_{b1} = [z \longmapsto 0]$

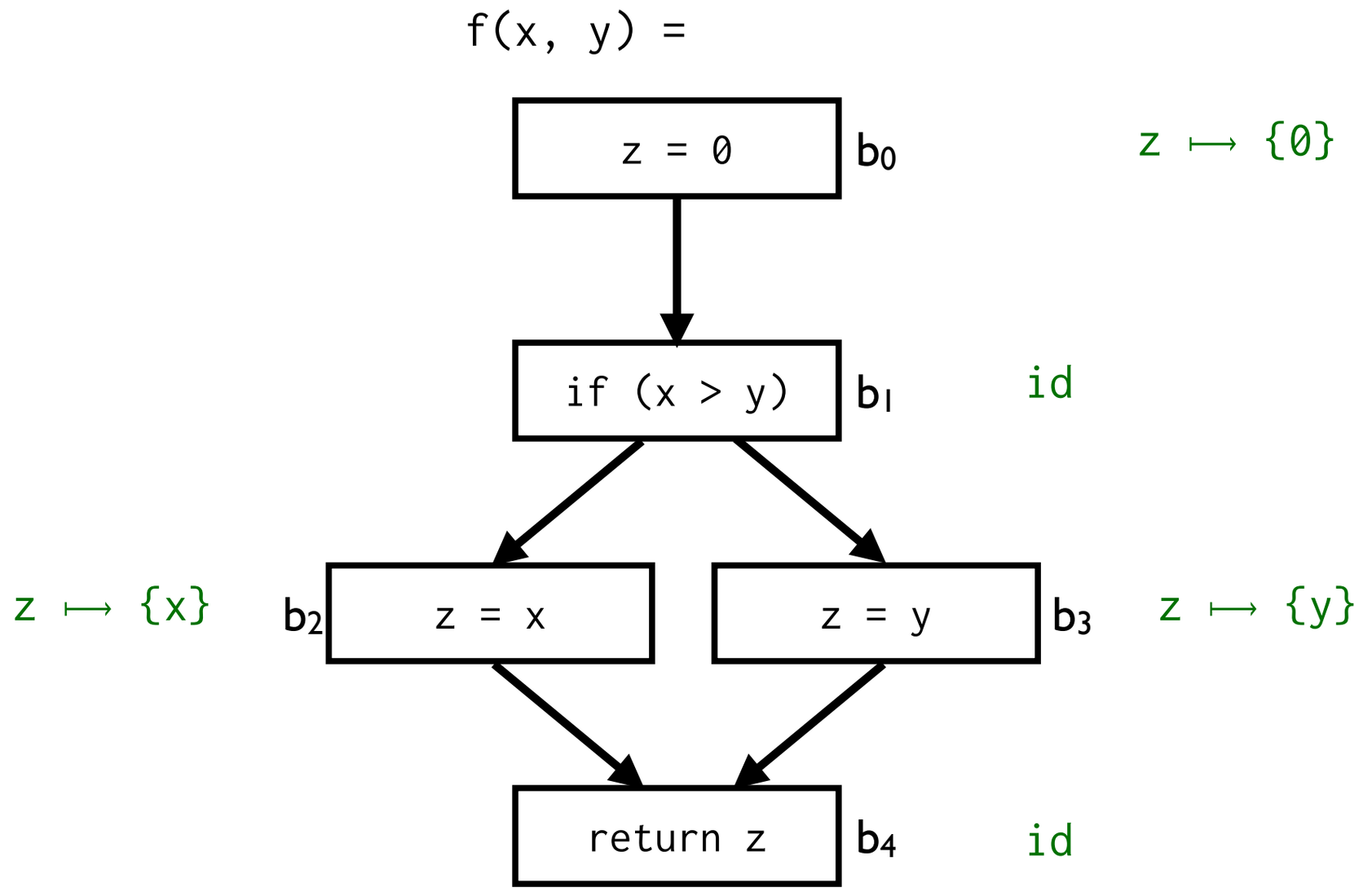
Composed Transfer Function



$trans_{b0} \circ trans_{b1} = [z \longmapsto 0]$

$trans_{b0} \circ trans_{b1} \circ trans_{b2} = [z \longmapsto \{x\}]$

Composed Transfer Function

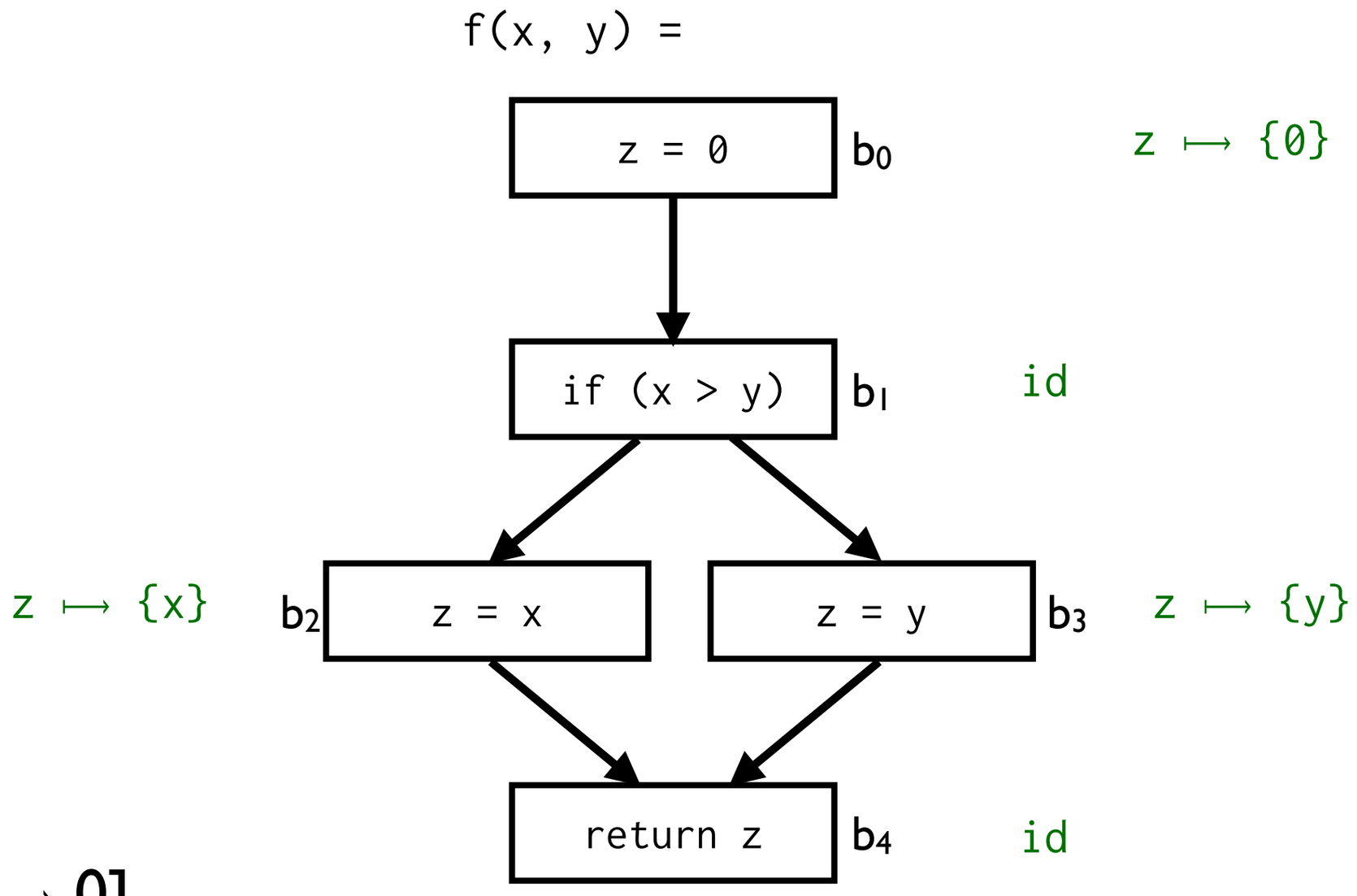


$trans_{b0} \circ trans_{b1} = [z \longmapsto 0]$

$trans_{b0} \circ trans_{b1} \circ trans_{b2} = [z \longmapsto \{x\}]$

$trans_{b0} \circ trans_{b1} \circ trans_{b3} = [z \longmapsto \{y\}]$

Composed Transfer Function



$trans_{b0} \circ trans_{b1} = [z \longmapsto 0]$

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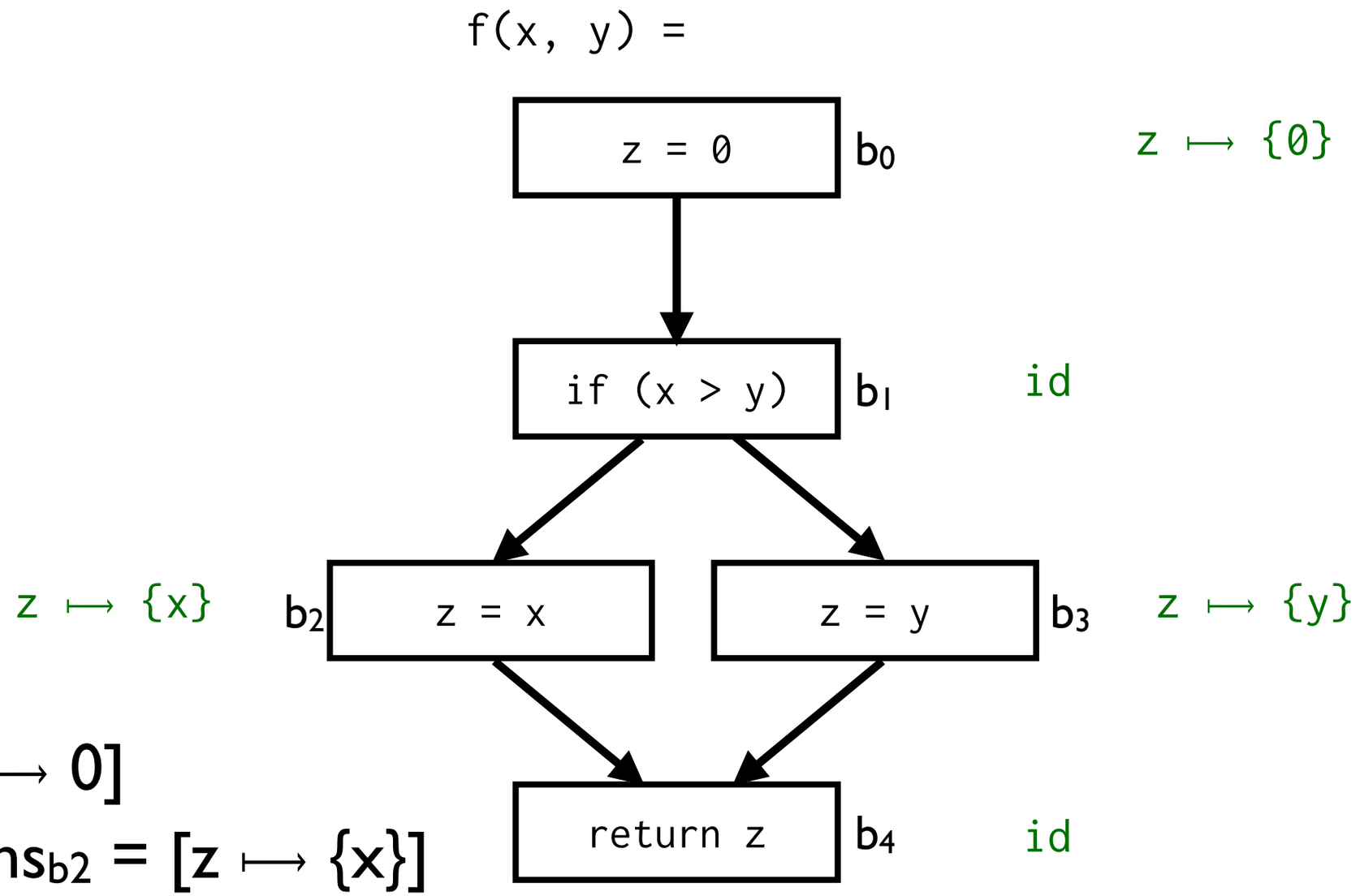
$trans_{b0} \circ trans_{b1} \circ trans_{b3} = [z \longmapsto \{y\}]$

$trans_{b0} \circ trans_{b1} \circ (trans_{b2} \sqcup trans_{b3}) = [z \longmapsto \{x,y\}]$

Composed Transfer Function



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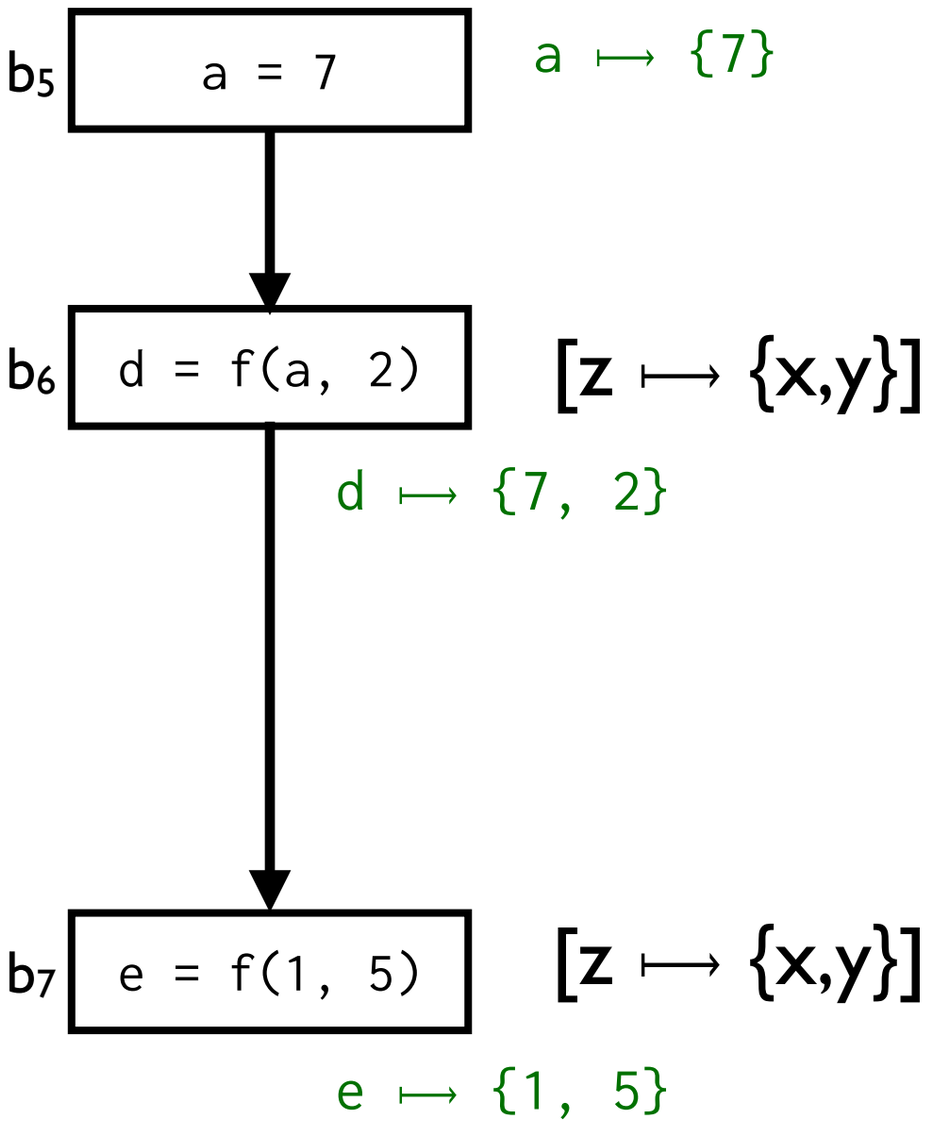
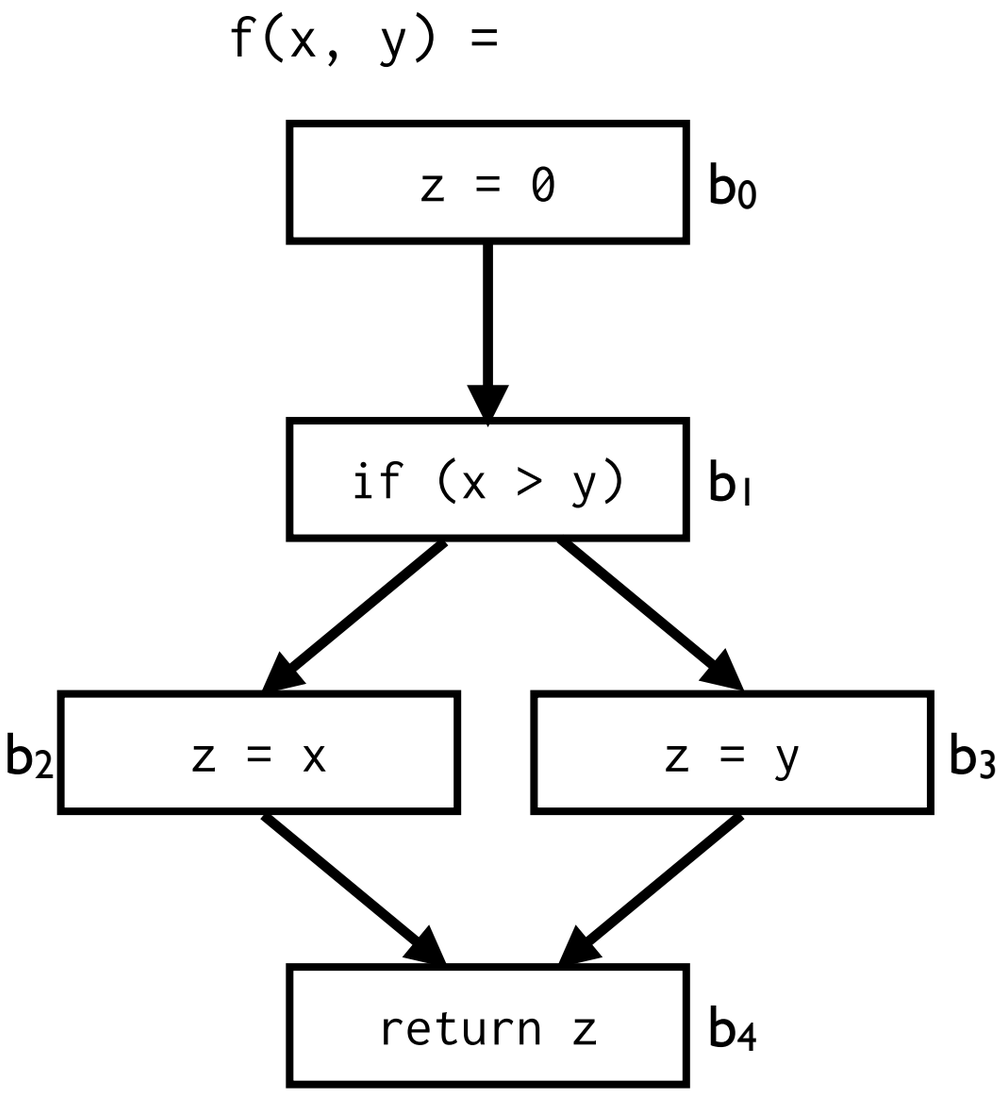
$trans_{b0} \circ trans_{b1} \circ (trans_{b2} \sqcup trans_{b3}) = [z \longmapsto \{x,y\}]$

$trans_{b0} \circ trans_{b1} \circ (trans_{b2} \sqcup trans_{b3}) \circ trans_{b4} = [z \longmapsto \{x,y\}]$

Context Sensitivity: Procedure Summaries





Composed Transfer Function

- Symbolically composing transfer functions yields a combined transfer function for f():
  - $trans_f$ = [`return` ⟼ {x,y}]
- Can use trans at each call site of f() without knowing the subroutine's body

Composed Transfer Function

- Limitations:
  - Requires suitable representation for symbolic result
  - Depends on strength of symbolic reasoning
  - Must analyze recursion patterns
  - Worst cases:
    - $trans_f = \top$ (i.e., as imprecise as intra-procedural analysis)
    - $trans_f$ is symbolic expression as complex as f itself

Heap Analysis

Limitations

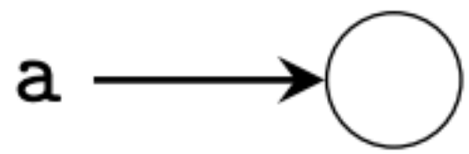
```
a = new A();
b = a;
c = new A();
a.x = 17;
c.x = 23;
b.x = 42;
print(a.x);
print(c.x);
```

- Program Analysis 1:
  - a.x and b.x and c.x are different things
  - We print 17 and 23
  - Unsound analysis!
- Program Analysis 2:
  - a and c and b might point to the same object
  - We print one of {17, 23, 42}, twice
  - Imprecise analysis
- Need to know:
  - Which variables can be aliases of each other?

Heap Graph

Example

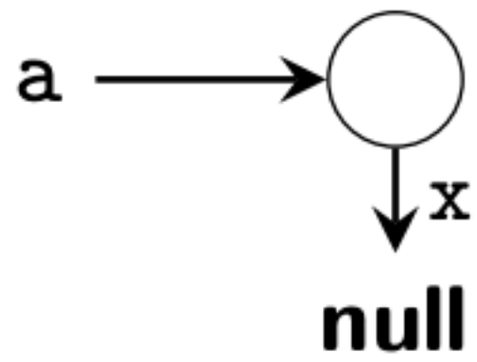
```
a := new();    // ⇐
a.x := null;
b := a;
b.x := new();
a.x.y := 1;
c := new();
c.x := new();
c.x.x := a;
c := a.x;
// A
```



Heap Graph

Example

```
a := new();
a.x := null;  // ⇐
b := a;
b.x := new();
a.x.y := 1;
c := new();
c.x := new();
c.x.x := a;
c := a.x;
// A
```



Heap Graph

Example

```
a := new();
a.x := null;
b := a;                    // ⇐
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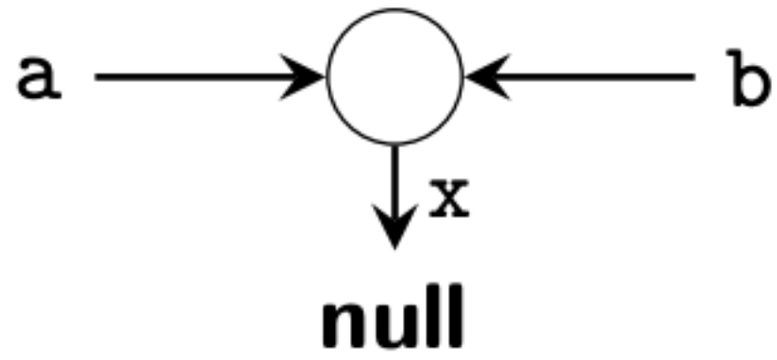
a.x.y := 1;
c := new();
c.x := new();
c.x.x := a;
c := a.x;
// A
```



Heap Graph

Example

```
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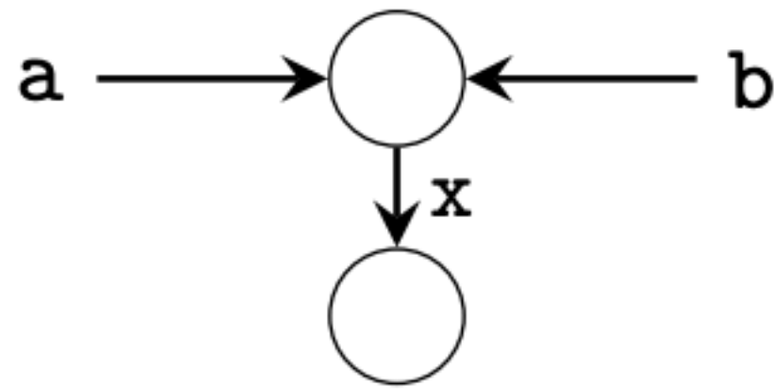
a.x := null;
b := a;
b.x := new(); // ⇐
a.x.y := 1;
c := new();
c.x := new();
c.x.x := a;
c := a.x;
// A
```



Heap Graph

Example

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a.x := null;
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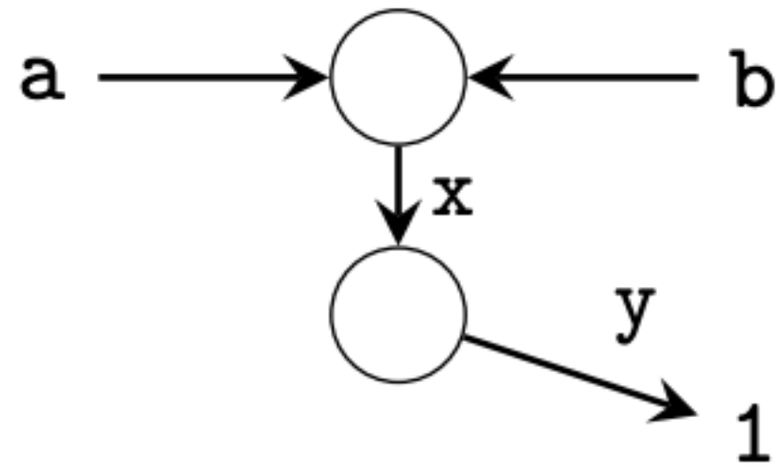
b.x := new();
a.x.y := 1;    // ⇐
c := new();
c.x := new();
c.x.x := a;
c := a.x;
// A
```



Heap Graph

Example

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b.x := new();
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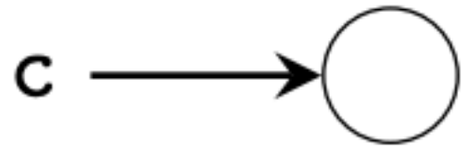
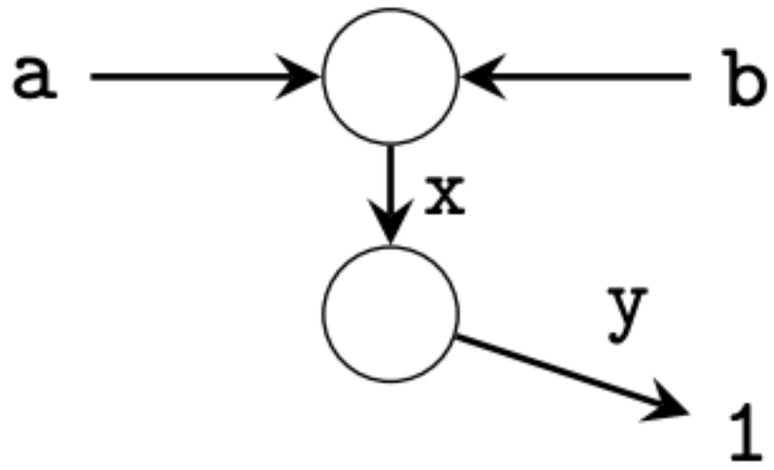
c := new();    // ⇐
c.x := new();
c.x.x := a;
c := a.x;
// A
```





Heap Graph

Example

```
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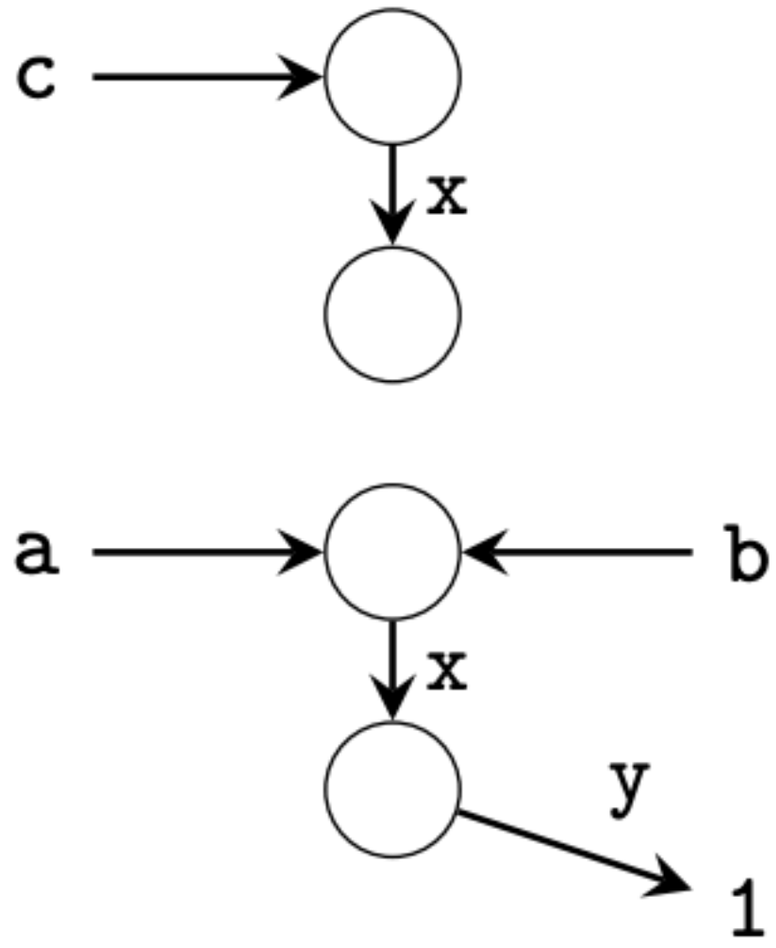
a.x := null;
b := a;
b.x := new();
a.x.y := 1;
c := new();
c.x := new(); // ⇐
c.x.x := a;
c := a.x;
// A
```



Heap Graph

Example

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a := new();
a.x := null;
b := a;
b.x := new();
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c.x := new();
c.x.x := a;    // ⇐
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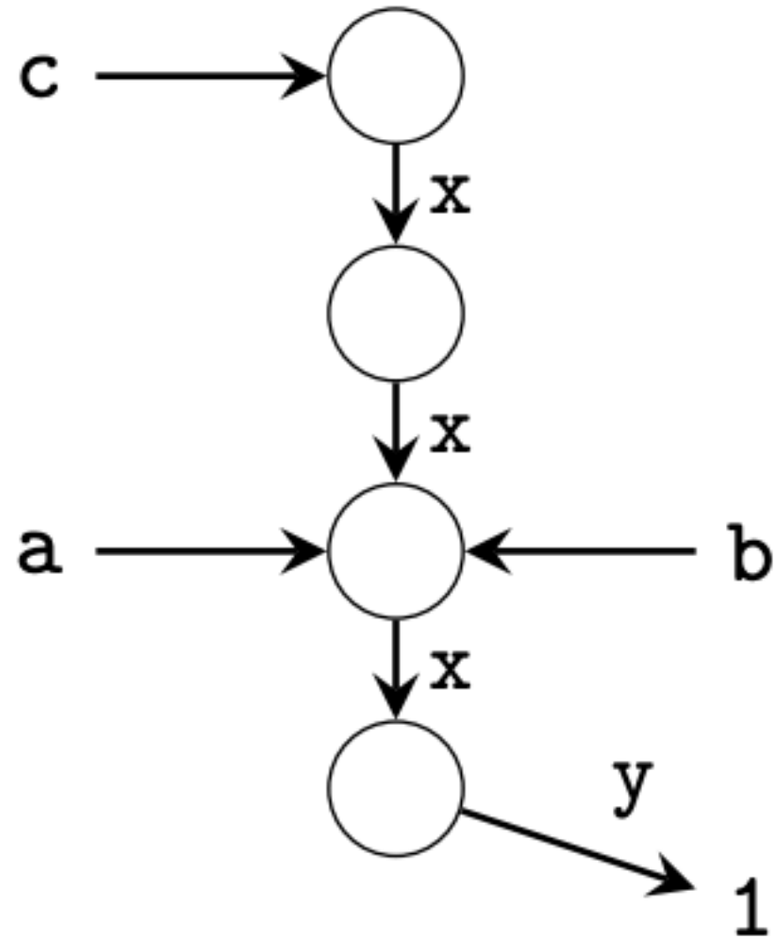
```



Heap Graph

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a.x := null;
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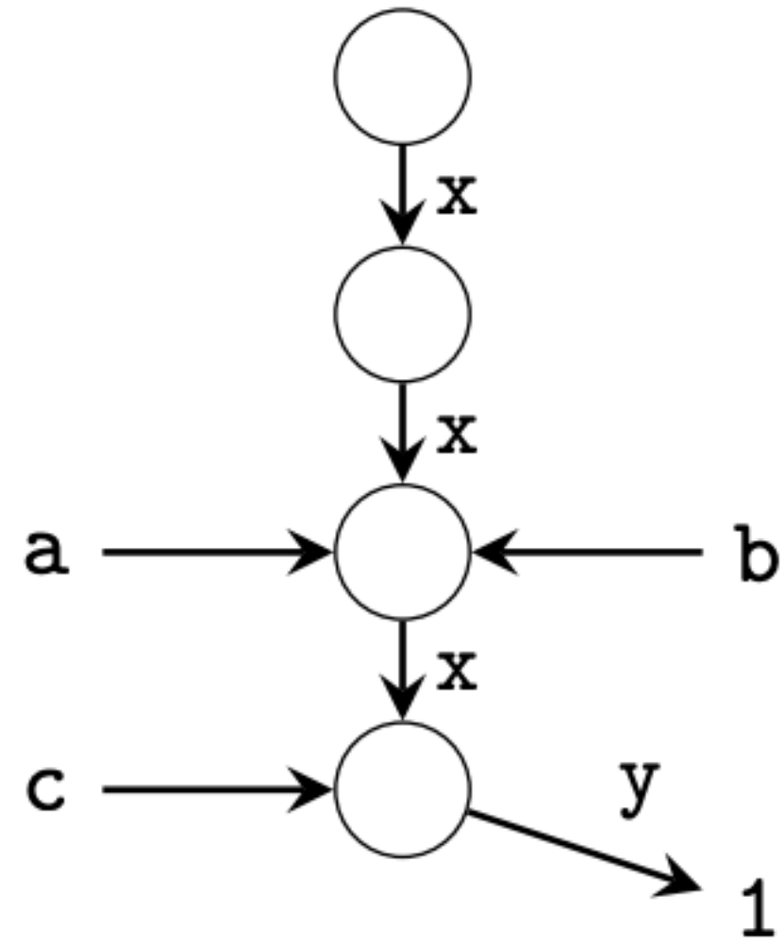
c.x.x := a;
c := a.x;      // ⇐
// A
```



Example

```
a := new();
a.x := null;
b := a;
b.x := new();
a.x.y := 1;
c := new();
c.x := new();
c.x.x := a;
c := a.x;
// A
```

Aliases at // A:

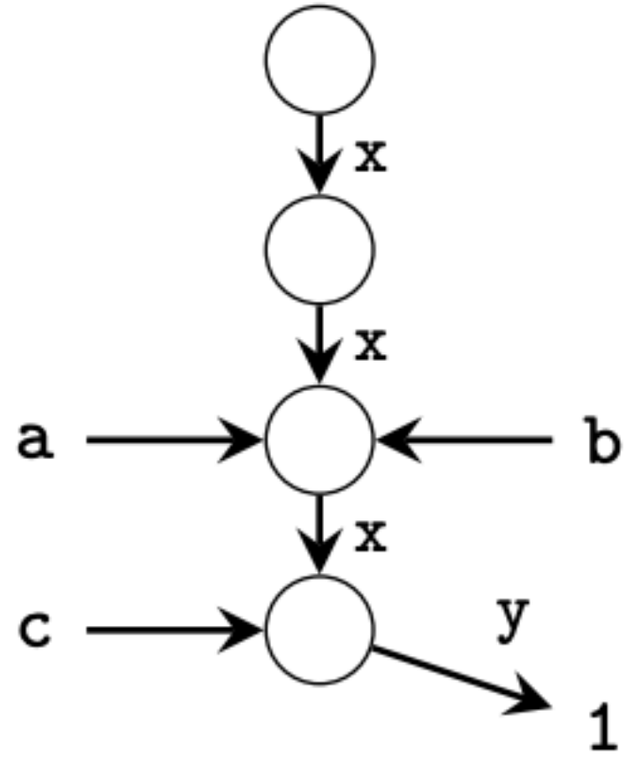
- a and b represent the same object

$\Rightarrow$ a and b are *aliased*

$$a \overset{alias}{=\!=} b$$

$\Rightarrow$ a.x and b.x are *aliased*

- c and a.x and b.x are *aliased*



Pointer Operations

Referencing	Dereferencing	Aliasing
C	C	C
`my_t *p = &var;` `p = malloc(8);`	`int x = *ptr;` `x = ptr2->field;`	`my_t *pa;` `pa = pb;`
Java	Java	Java
`A a = new A();`	`int x = a.f;`	`A b = a;`

Points-to Information

- Variable pairs that refer to the same memory location
  - <*a, b>, <*c, b>, <*a, *c>
  - *a and b alias, same with *c and b
- Points-to pairs:
  - <a → b>, <c → b>
  - a points to b, and c points to b (hence *a and *c are alias)
- Alias sets:
  - {*a, b, *c}
  - They all point to the same memory location

Locations

- Static locations
  - Uniquely identified by name
- Stack-dynamic locations (Local vars, parameters)
  - Named
  - May never exist (function not called)
  - May exist more than once (recursion)
- Heap-dynamic locations
  - Not named
  - May exist arbitrarily often

Heap/Stack Locations: Design Space

- Don't distinguish, or
- Distinguish by:
  - Type
  - Name
  - Calling context (Context-sensitive analysis)
  - Control flow (Flow-sensitive analysis)
  - Containing heap objects (Object sensitivity)
  - Referencing field name in objects (field sensitivity)
  - Array index (array sensitivity)
  - Pointer arithmetic
  - Literature has explored many options

Steensgaard's Points-To Analysis

- Steensgaard, B. (1996, January). Points-to analysis in almost linear time. In Proceedings of the 23rd ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages (pp. 32-41).

- Fast: $O(n\alpha(n,n))$ over variables in program
- Developed to deal with large code bases at AT&T
- Sacrifices Precision
- Equality-based
- Basis for many more precise algorithms

Steensgaard's Points-To Analysis

- C pointer semantics:
  - &a: Address of a
  - *a: Object pointed to by a
  - Converse operators: *(&a) = a
- Steensgaards analysis considers four cases:

	C	Java	
Referencing	`a = &b`	`a = new A()`	$\ell_b \in$ pts(a)
Aliasing	`a = b`	`a = b`	pts(a) = pts(b)
Dereferencing read	`a = *b`	`a = b.f`	for each $\ell \in$ pts(b) → pts(a) = pts(ℓ)
Dereferencing write	`*a = b`	`a.f = b`	for each $\ell \in$ pts(a) → pts(b) = pts(ℓ)

Steensgaard's Points-To Analysis

```
int i, j, k;
int *a = &i;
int *b = &k;
a = &j;
int **p = &a;
int **q = &b;
p = q;
int *c = *q;
```

Actual:

Steensgaard:

Steensgaard's Points-To Analysis

```
int i, j, k;
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Actual:

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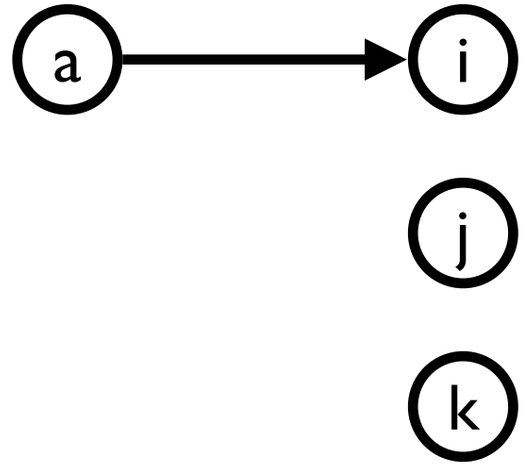

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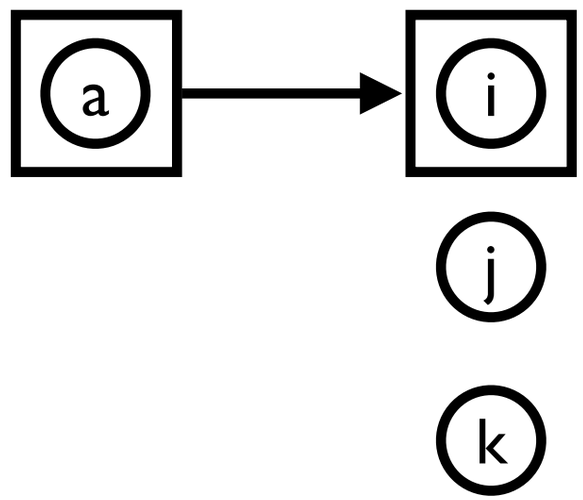

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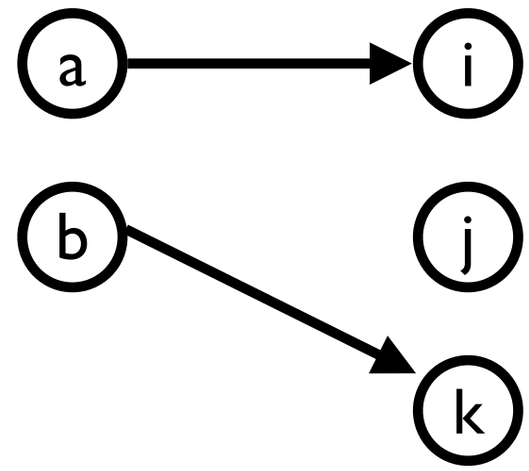

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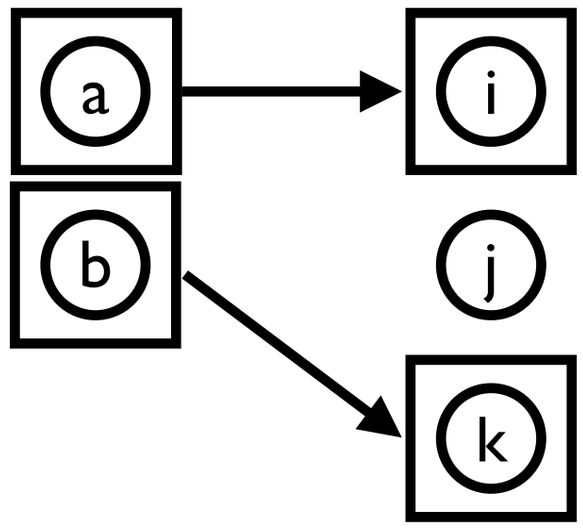

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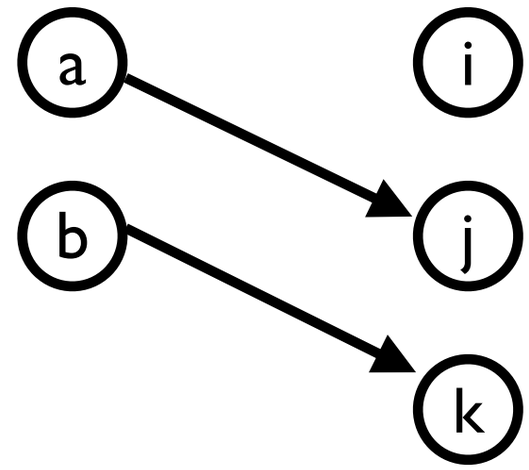
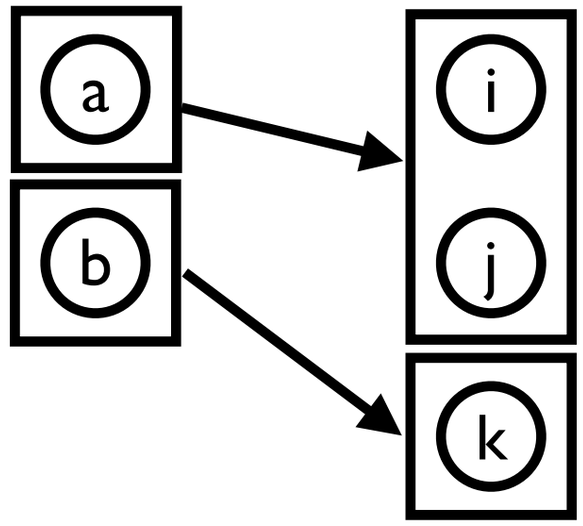


Steensgaard:



Steensgaard's Points-To Analysis

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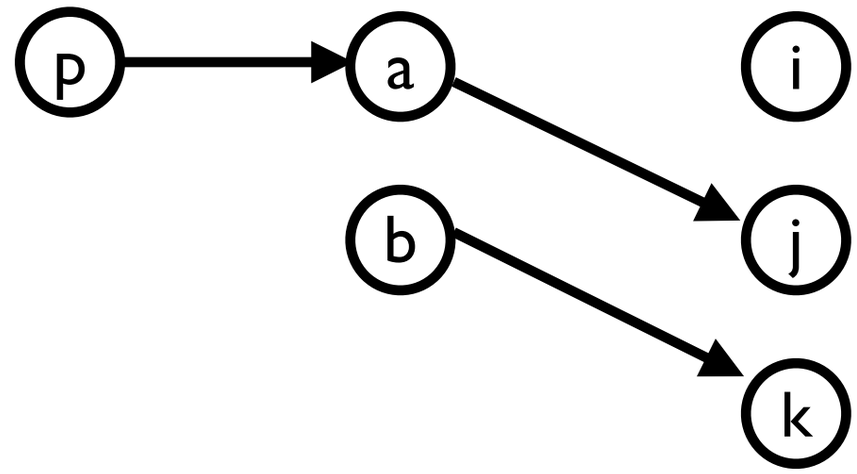
Actual:



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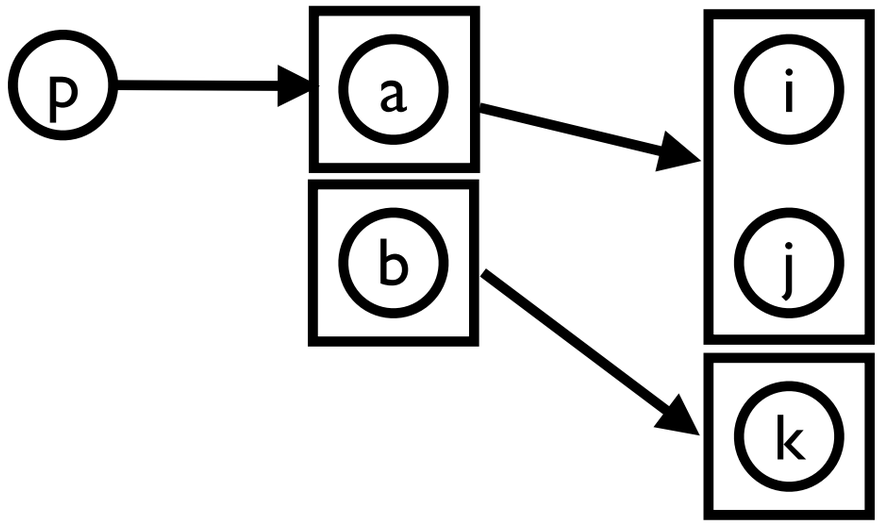

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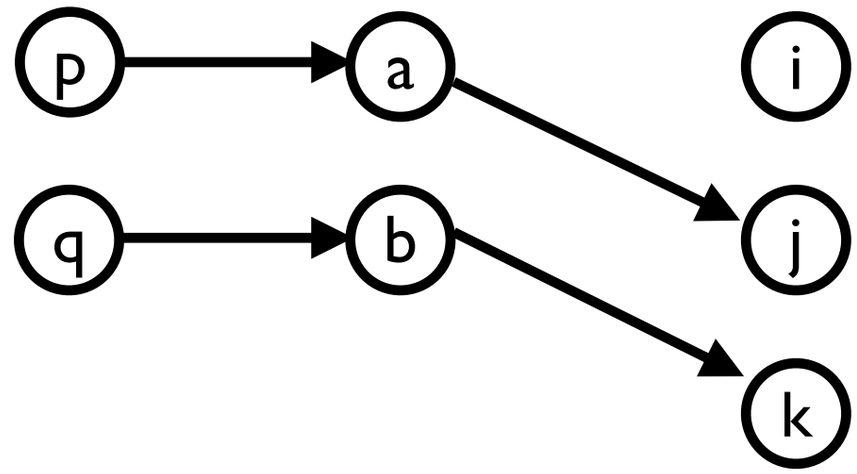

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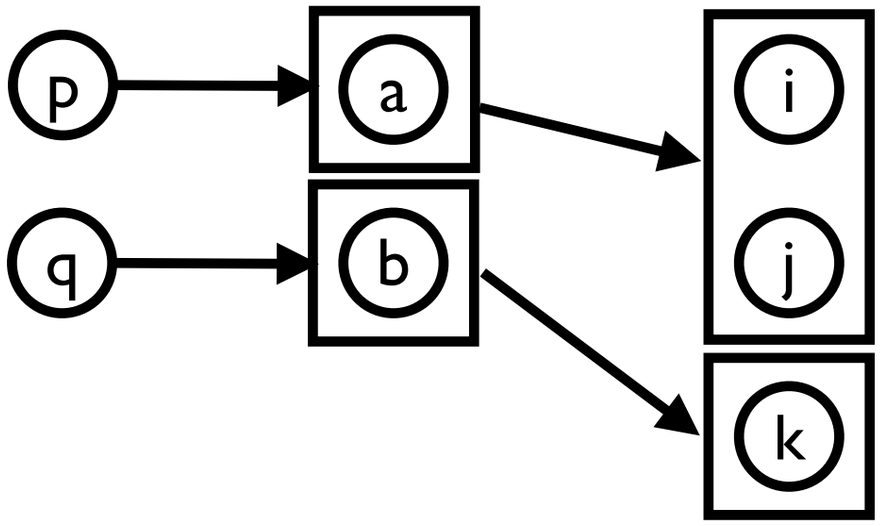

Steensgaard's Points-To Analysis

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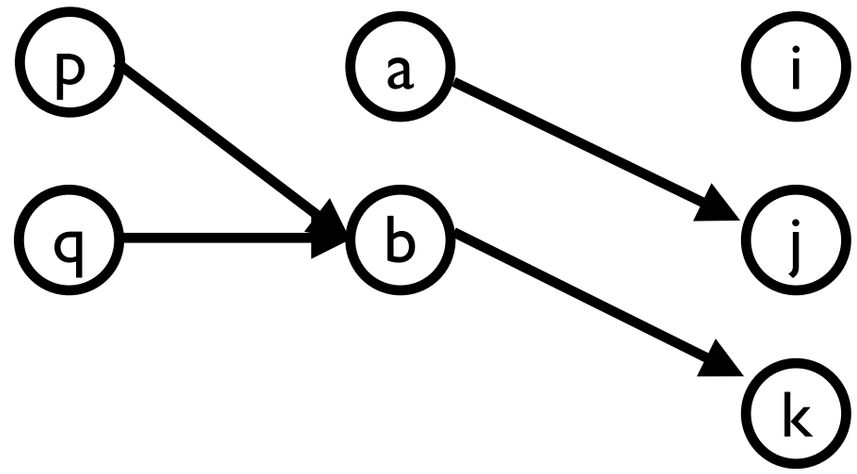

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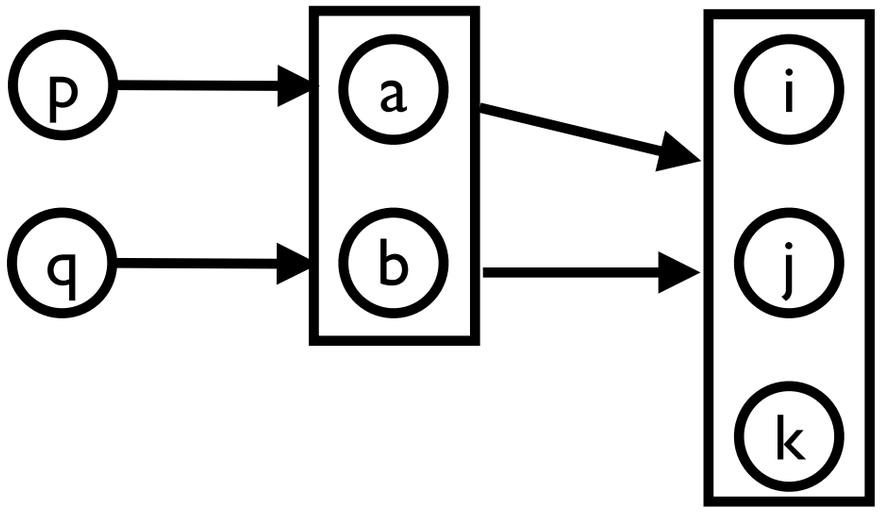

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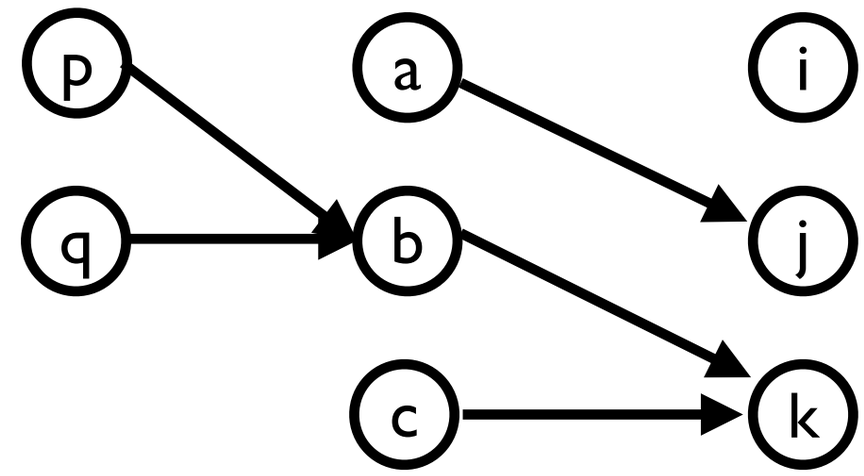

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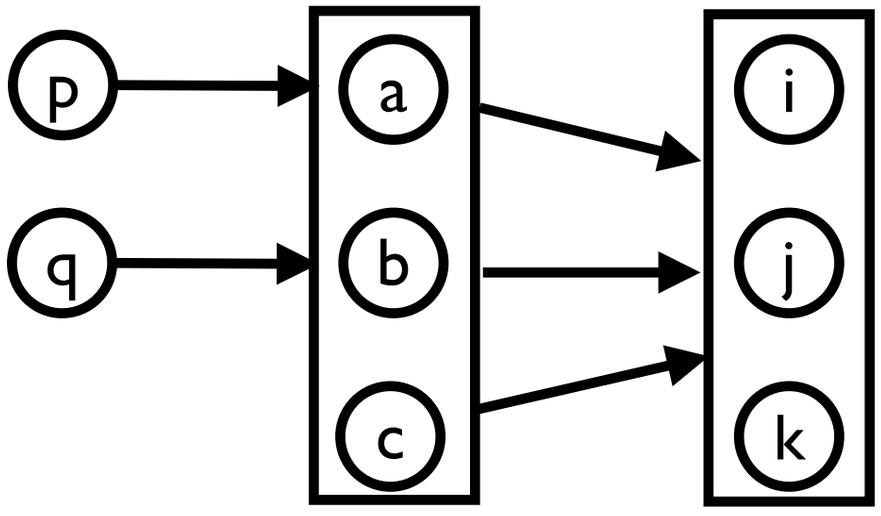

Steensgaard's Points-To Analysis

- Points-to sets pts(v) serve as abstraction over addresses that v can point to
- Steensgaard's points-to-analysis:
  - Insensitive to flow, context, fields, …
- Efficiently implemented using Union-Find data structure
- Highly efficient
- Imprecise

Andersen's Points-To Analysis

- Andersen, L. O. (1994). Program analysis and specialization for the C programming language (Doctoral dissertation, University of Copenhagen).

- Asymptotic performance is $O(n^3)$

- More precise than Steensgaard's analysis
  - Still not terribly precise

- Subset-based (a.k.a. inclusion-based)

- Basis for many algorithms with performance improvements

Andersen's Points-To Analysis

- Collect constraints, resolve as needed
- For each statement in program, we record:
  - If Referencing (a = &b):
    $\{\ell_b\} \subseteq$ pts(a)
  - If Aliasing (a = b):
    pts(b) ⊆ pts(a)
  - If Dereferencing read (a = *b):
    pts(*b) ⊆ pts(a)
  - If Dereferencing write (*a = b):
    pts(b) ⊆ pts(*a)

Andersen's Points-To Analysis

Actual:

```
int i, j, k;
int *a = &i;
int *b = &k;
a = &j;
int **p = &a;
int **q = &b;
p = q;
int *c = *q;
```

Andersen:

Andersen's Points-To Analysis

```
int i, j, k;
int *a = &i;
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int *c = *q;
```

Actual:

Andersen:

Andersen's Points-To Analysis

```
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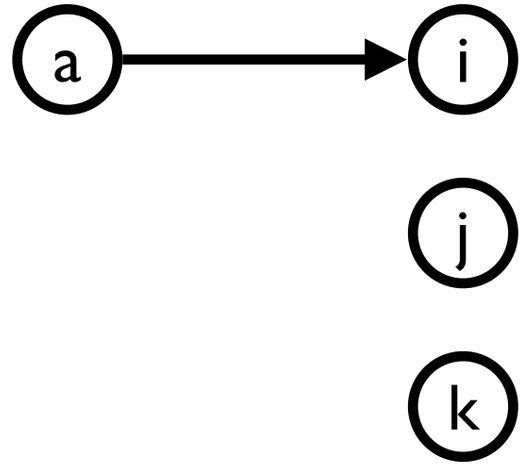
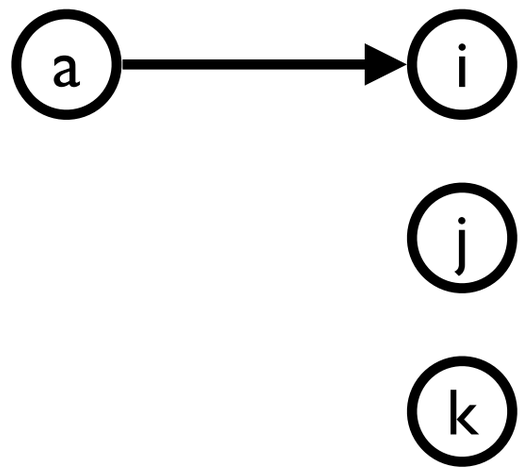
```

Actual:



Andersen:



Andersen's Points-To Analysis

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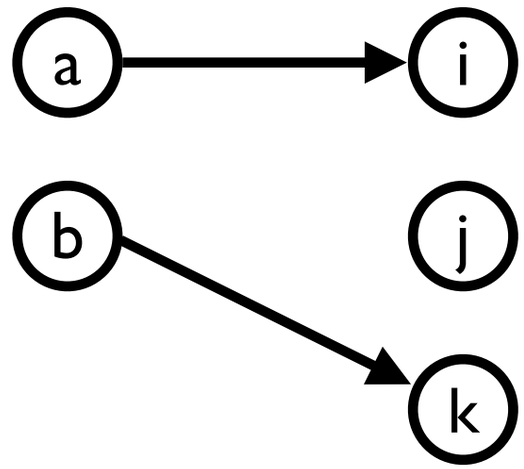
Actual:



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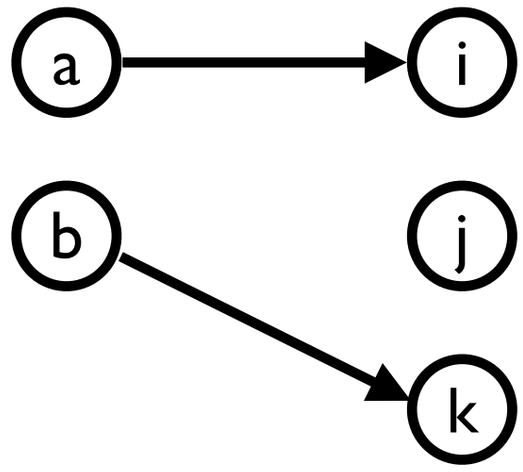

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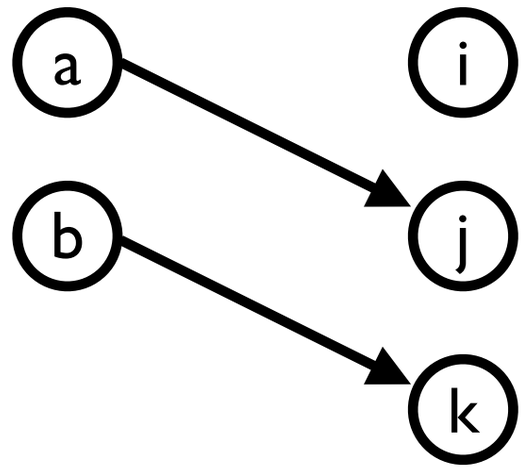

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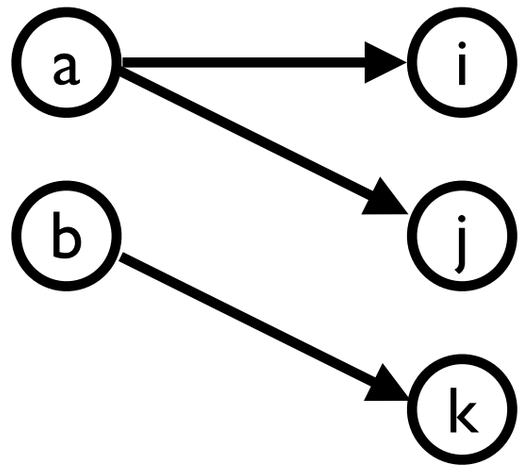

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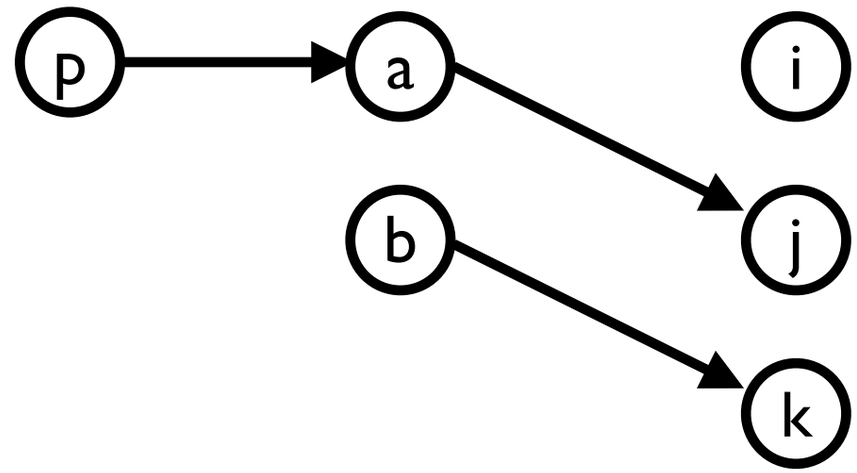

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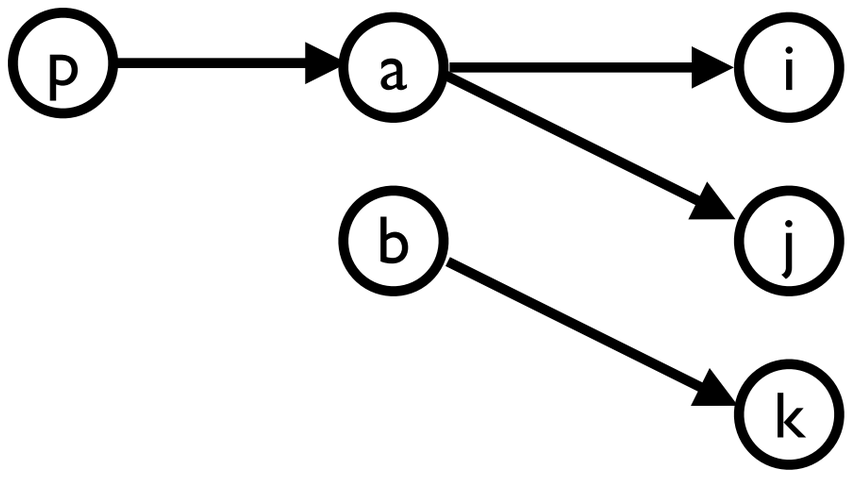

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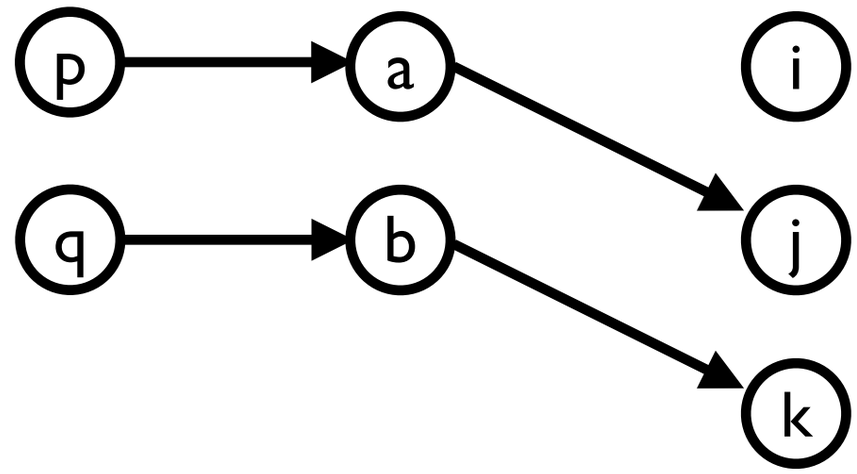

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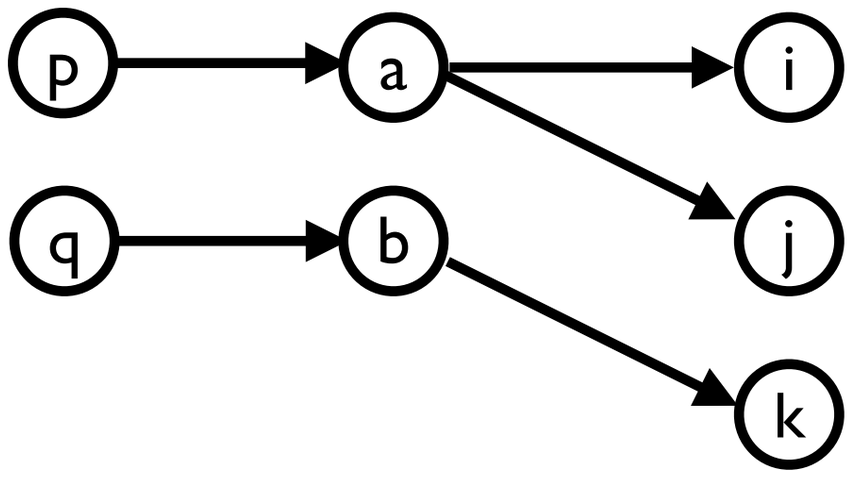

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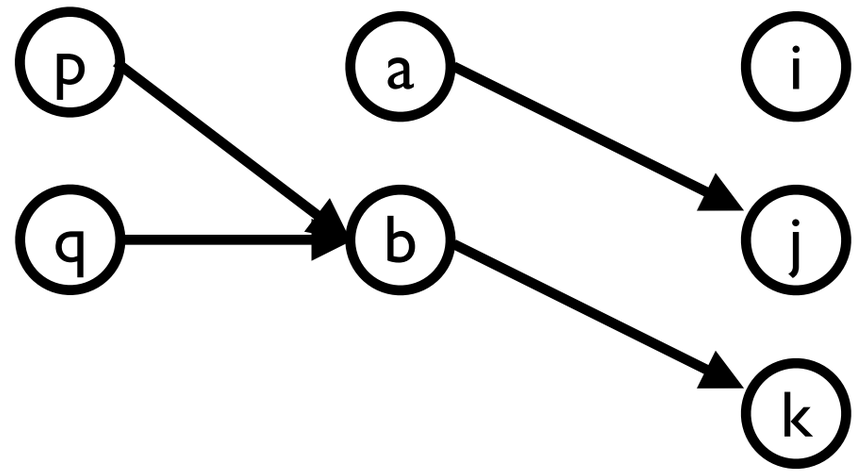

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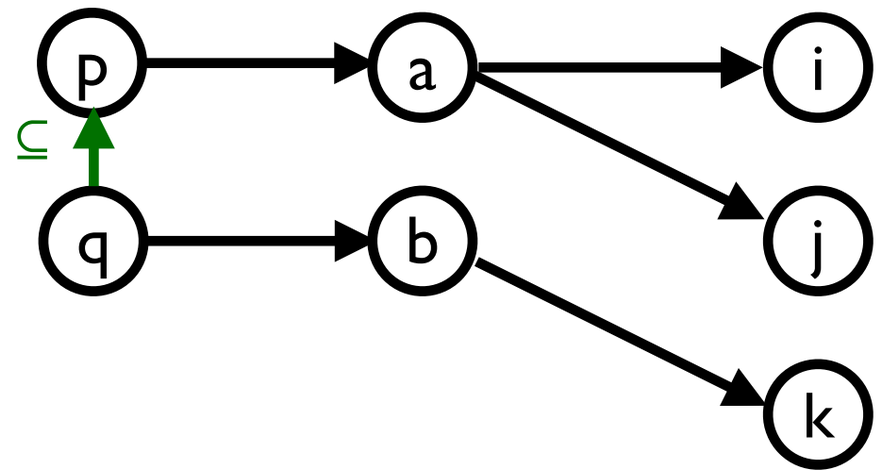

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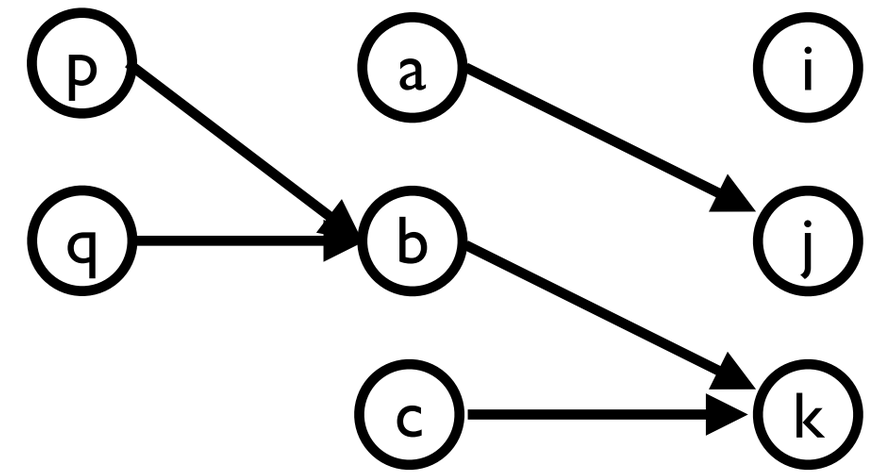

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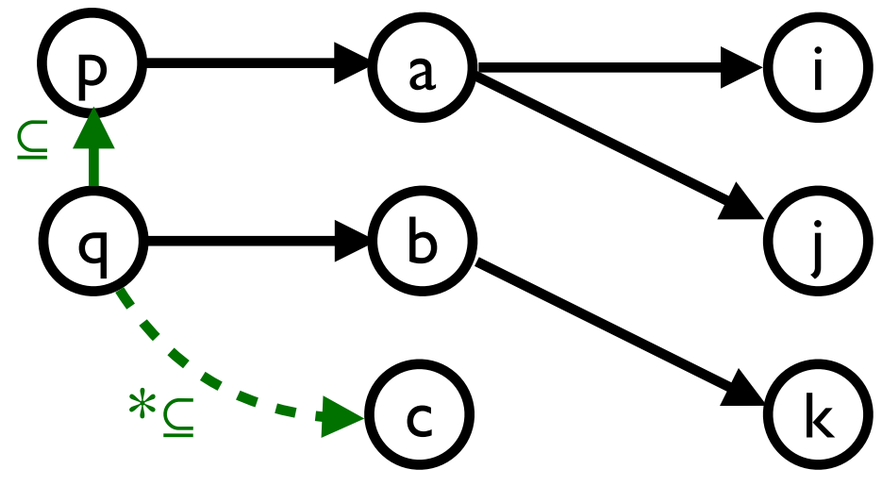

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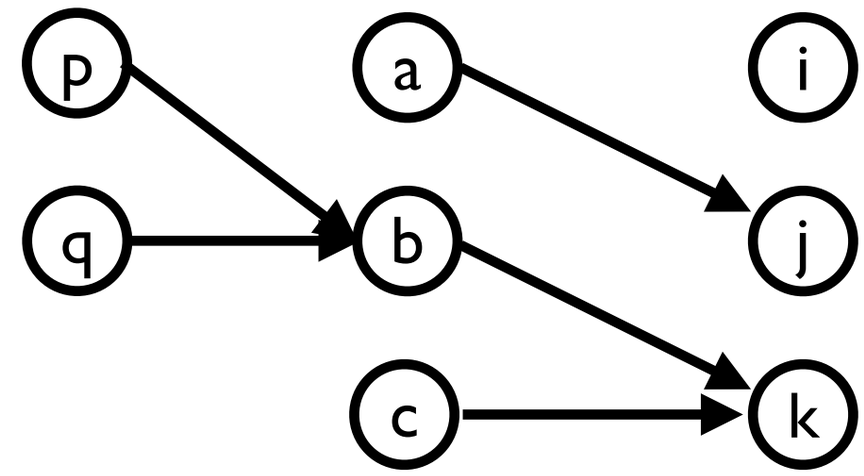

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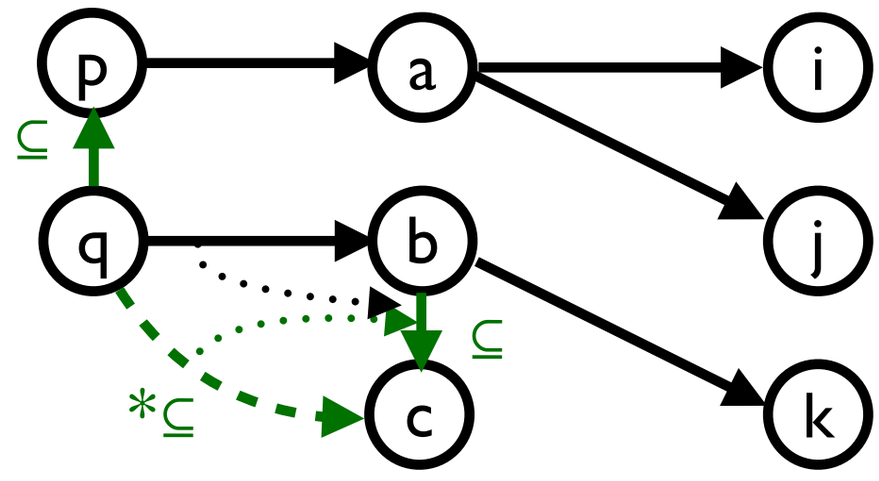

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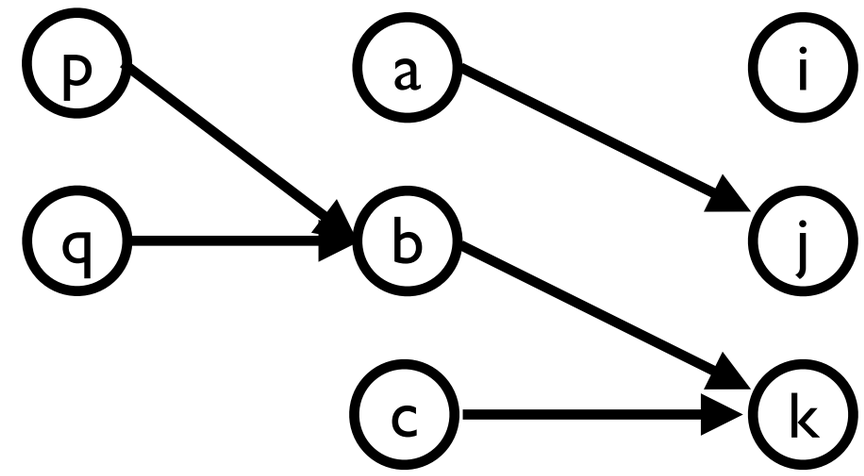

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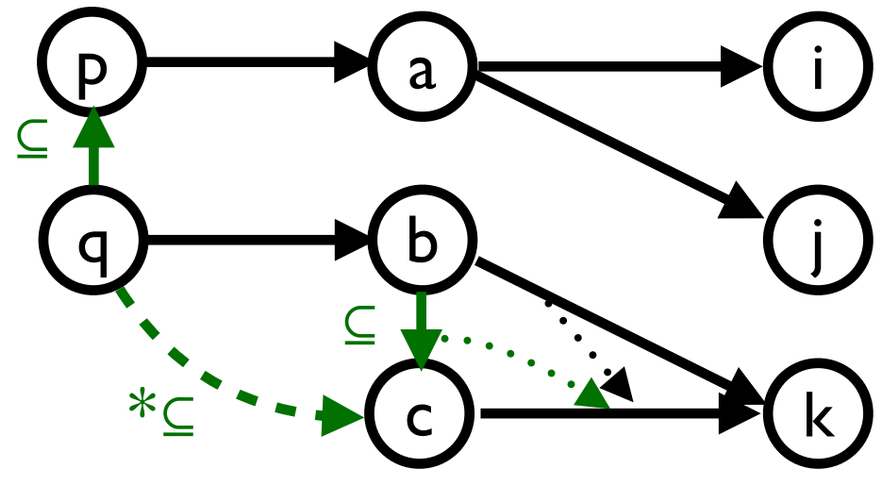

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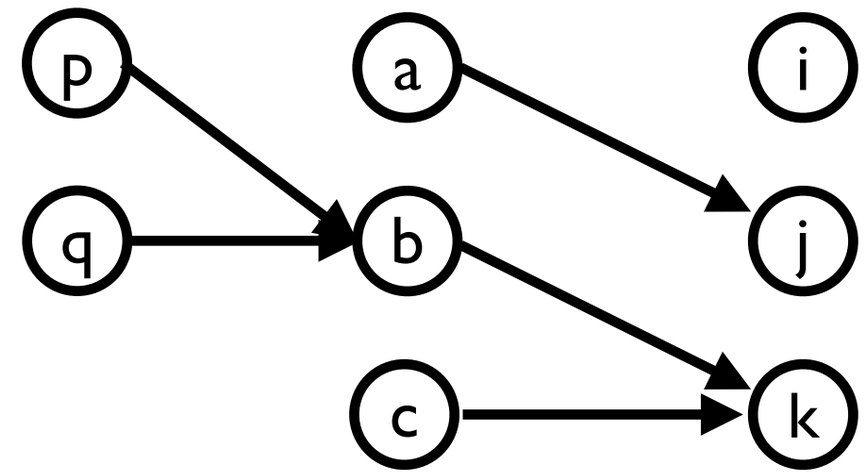

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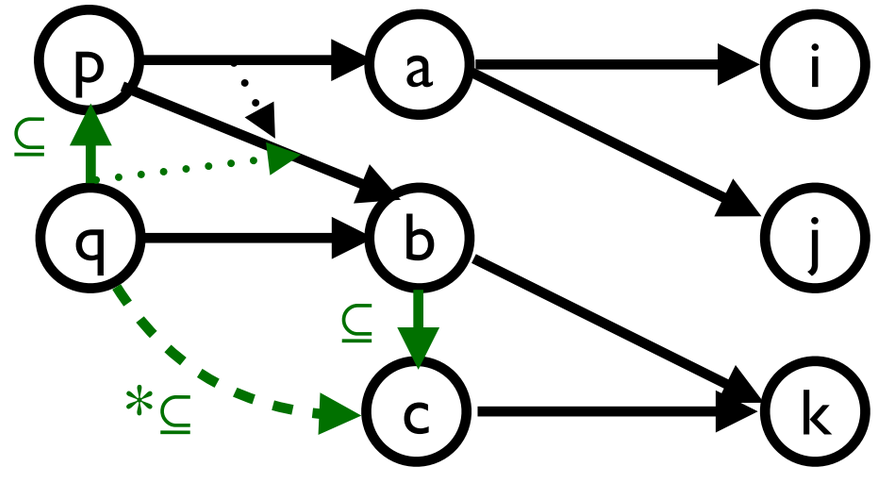

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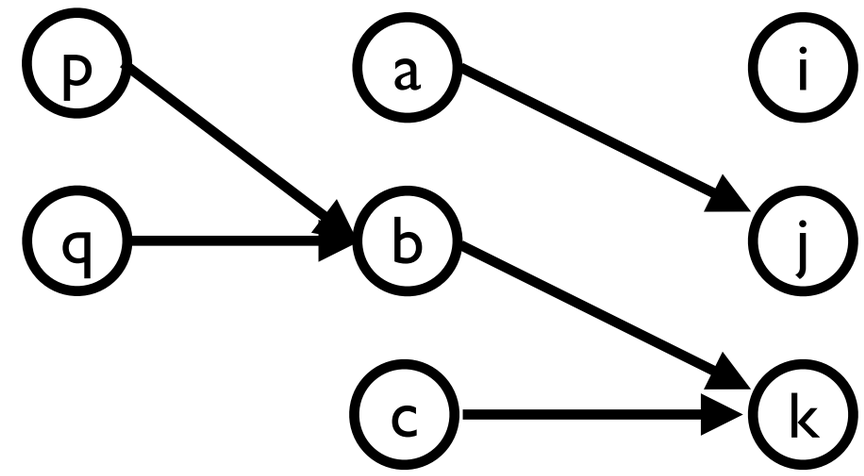

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